Less Paper, More Public Service: Communication Technology inside the State *

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November 2, 2026

Job Market Paper

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Abstract

Public services often leave capacity unused while demand goes unmet. These gaps can persist because the decisions that would close them require information to travel between state entities, often as paper documents. I exploit the staggered rollout of a digital platform in Chile that replaced paper in official communication between state entities. Within the national public childcare agency, adoption cuts document processing times by 70%, and documents from the departments that coordinate enrollment rise, concentrated in the months when seats are assigned. Enrollment at the agency's centers rises by 11%, with no increase in capacity, and no detectable change in applications per eligible child or at providers outside the reform. The increase is concentrated in centers whose enrollment relies more on these documents. The pattern is consistent with better communication between state entities raising public output.

Keywords: Organizational economics, state capacity, public administration, digital transformation.
JEL Classification: D23, H11, L86, O38.

*I am deeply grateful to my supervisors, Pedro Souza and Lucie Gadenne, and to Jonas Hjort and Felipe Gonzalez, for their invaluable guidance and insights. I also thank Camila Arroyo, Nicolas Castro, Diego Dabed, Antonio Leon, Laura Perez and Joaquin Urria, as well as discussants and participants at the EEA-ESEM 2026 Congress, QMUL internal seminars, the QMUL PhD Conference and the 30th Annual SIOE Conference, for their valuable comments. All errors are my own.

1 Introduction

What drives efficiency in the delivery of public services? A natural answer is more resources, but they are often scarcest where they are needed most (e.g., Besley & Persson, 2009). Part of the answer, however, requires no new ones (Best *et al.*, 2023). Public services often leave capacity unused while demand remains unmet: acute hospital beds sit empty while patients wait months for treatment (OECD, 2025), and schools turn children away while others have seats to spare (OECD, 2018). Some of this unused capacity is inefficient.[1] Matching it to those who wait takes several entities of the state acting in sequence. Their decisions depend on one another (Besley *et al.*, 2022), and each decides with information the others hold. They exchange this information mainly through formal documents, often on paper. Thus, better communication could help the state allocate the capacity it already has.

Digital platforms transform this exchange. Since 2005, governments have been moving official communication onto digital platforms.[2] This paper studies the case of Chile, where, as in Italy, Germany, and Japan, official communication has long relied on signed and stamped paper documents.[3] In 2019, a law mandated that official communications between state entities be registered on a centralized platform.[4] The technology changed how information moves, not what it says, who decides, or the steps that produce a document.

I study how the technology affected communication in Chile's National Board of Kindergartens (henceforth, the agency) and how enrollment responded in the childcare centers it oversees. This setting is ideal for three reasons. First, placing a child is a

[1]The rest can be efficient, as insurance against emergencies or because it lacks complementary inputs, such as staff.

[2]See, for example, Estonia's document exchange center in 2005 (State Chancellery of the Republic of Estonia, 2005), Korea's On-nara in 2006 (Moon, 2019), Brazil's SEI in 2015 (Presidência da República, Brasil, 2015), Argentina's Gestión Documental Electrónica in 2016 (Presidencia de la Nación, Argentina, 2016), and Peru's Modelo de Gestión Documental in 2017 (Secretaría de Gobierno Digital, Perú, 2017). No cross-country survey records their diffusion.

[3]For example, written form remains mandatory under most German federal administrative rules (Deutscher Bundestag, 2020), and as of 2020 most Japanese national procedures could not be completed online (Cabinet Secretariat IT Strategy Office and Ministry of Internal Affairs and Communications, Japan, 2020).

[4]Law No. 21.180, the *Ley de Transformación Digital del Estado*. The platform it created is *DocDigital*. Section 2.3 describes both.

task that no single entity can complete on its own. The national directorate coordinates enrollment, and families apply to one center. The regional directorate (i.e., the agency's regional office) ranks the applicants to each center and sends it the list, and each center records, or reports to the directorate, whether each family takes the seat. A seat stays attached to the child it is offered to until the outcome is recorded, and only then is it freed and offered to the next child on the waiting list (JUNJI, 2022b). Second, the regional directorate communicates with all its centers through the formal documents the reform digitized. These carry general guidelines on management and services, and the rules and follow-ups of the enrollment process. For some centers, the reform also reached the enrollment task itself. Directly managed centers record enrollment on the agency's enrollment system, which the reform left untouched. Indirectly managed centers, run by third parties, do not all have access to it. Those outside the system send their enrollment forms to the directorate as documents, and the directorate enters them into the system and sends the results back the same way (JUNJI, 2022a). Comparing the two types of centers allows me to separate communication in general from the communication the task relies on. Finally, I observe both communication and enrollment, every document the agency sent between 2017 and 2023 and enrollment for every center under each regional directorate, so I can relate one to the other.[5]

Public childcare matters economically, and seat allocation is a recurring challenge. Early-childhood education has large, persistent effects on children and their families (e.g., Chetty *et al.*, 2011; Almond *et al.*, 2018; Andresen & Havnes, 2019; Gray-Lobe *et al.*, 2023). The agency accounts for 53% of Chile's preschool enrollment (Arias *et al.*, 2024). In 2019, more than one in four applicants received no placement. At the same time, nearly half of the centers with a waiting list also had empty seats: of every five children waiting there, two could have been seated without adding a seat.[6]

Identification uses the staggered rollout of the platform between 2020 and 2023 for

[5]The agency's registry records every formal communication it sends, before and after adoption, 146,170 documents in the estimation sample, each classified by sender department and receiver and with the center's management type. The classification is by the department's function, not by document content, which the registry does not record.

[6]Of 136,801 applicants, 37,746 received no placement. Of the 1,620 centers with a waiting list, 766 also had empty seats: 12,881 children waiting and 5,557 seats empty in those same centers. These counts are stocks.

both outcomes, communication and enrollment.[7] The parallel-trends assumption holds: early and late adopters follow the same pre-adoption trajectories in both outcomes. For communication, the unit is the sender–receiver link. The platform does not require both sides to adopt, since a sender that has adopted dispatches its documents digitally even if the receiver has not. Adoption by either side can therefore change how the documents on a link are produced or how many are sent, so a link is treated from its first adoption. For both processing time and document counts, I compare the evolution of each link before and after that first adoption against links where neither side has adopted yet.[8]

I first show that adoption changed the speed and composition of the agency's communication. Processing time drops by 70% compared to the pre-adoption average of 1.1 working days, before counting the transit that digital dispatch removes.[9] Documents sent by the departments responsible for coordinating enrollment increase by 34%, concentrated in the October-to-March months when the agency assigns seats for the next academic year. Contact through these documents also becomes more frequent, with more weeks in which a pair sends at least one. The total number of documents shows no statistically significant change.[10]

Turning to enrollment, the reform raises the number of children enrolled without adding capacity. A center is treated when the regional directorate that sends it documents adopts. I compare centers before and after adoption within the same group of regions and year.[11] Enrollment at the agency's centers rises by 4.2 children per year, 11% of its pre-adoption level, with no new centers, no rise in capacity, and no statistically significant change in applications per eligible child. As a placebo, enrollment at providers outside the agency, which the reform did not reach, shows no statistically significant

[7] Adoption timing was set by the deployment of the central digital-government office rather than by the entities themselves.

[8] The specification includes link and time fixed effects, which absorb administrative seasonality, and a directorate-specific trend.

[9] Processing time is the number of working days from the request for a document number to the dispatch of the document, measured on the documents the agency sends. It does not include delivery to the receiver. For example, the national directorate routinely communicates with the regional directorates, on average 772 km away and up to 2,193 km.

[10] The confidence interval rules out increases above 7% of the pre-adoption average.

[11] The specification includes center fixed effects and time effects common to each macrozone, a group of neighboring regions.

change.

The increase is concentrated in indirectly managed centers, where enrollment relies more on documents. In these centers, enrollment rises by about 5.3 children per center, while directly managed centers show no statistically significant increase. Across the 1,690 indirectly managed centers, this means about 9,000 additional children each year for a given level of seats.

The evidence suggests that digital communication complements the tasks that rely on it. Both types of centers receive the same general communication from the directorate, so the differential effect points to the documents that only indirectly managed centers exchange as part of enrollment. In these centers, more offers are made over the year, families accept them at the same rate, and more children enroll. Directly managed centers show no statistically significant change in offers or enrollment. This pattern is consistent with a document-dependent handoff, in which the directorate records each outcome and returns the results to these centers as documents. A limitation is that I cannot tell which step of the admission process changed, since I observe only the documents and the enrollment records.

The effect on enrollment is progressive and benefits children from lower-income families. In indirectly managed centers, three in four of the additional children come from the poorest 60% of households. What these children cost depends on whether staff grew with them. It ranges from about US$1.4 million a year, if they filled groups that already had staff, to about US$19.8 million, if staff grew in proportion to enrollment.[12]

I consider four alternative explanations. First, digitization may have freed administrative staff time at the center. The increase is not statistically larger in directly managed centers with no administrative staff, or in indirectly managed centers too small for the agency to fund an administrative assistant.[13] Second, families, not the agency, may drive the increase. Applications per eligible child, the acceptance rate, and written queries

[12]Both scenarios use 2019 costs for the roughly 9,000 additional children. The first includes only the inputs that scale with a child, about US$160 per child-year. The second includes the full running cost of a seat in an indirectly managed center, about US$2,200 per child-year. The estimate of 9,000 children has a wide interval, from about 3,600 to 16,700.

[13]If digitization freed staff time, the increase should be larger where administrative capacity is binding. The agency funds an administrative assistant only in indirectly managed centers with a capacity of 100 seats or more (JUNJI, 2019).

to the directorate show no statistically significant change.[14] Third, the increase could be a transfer of children from directly managed centers, but enrollment at those centers shows no decline, and enrollment across the agency rises. Fourth, indirectly managed centers are paid per child attending, but the effect is similar in a year when payment did not depend on attendance.

Overall, the paper has three main findings. First, the technology cut the time the agency takes to process its documents and increased the documents it sends to coordinate enrollment. Second, the agency enrolled more children without expanding its capacity. Third, the increase is concentrated in indirectly managed centers, a pattern consistent with their greater reliance on documents. The findings suggest that better communication between the entities of the state raises delivery through the tasks that rely on it, and so improves how the state uses its capacity.

The paper contributes to three literatures. The literature on state effectiveness has emphasized personnel, incentives, and authority as sources of state performance (e.g., Khan *et al.*, 2016; Finan *et al.*, 2017; Bandiera *et al.*, 2021; Best *et al.*, 2023; Muñoz & Otero, 2025). These studies treat each public entity as an organization. The state as a whole is also one, whose entities depend on one another, and this level has received little empirical attention (Besley *et al.*, 2022). Closest to this paper, Bandiera *et al.* (2021) shift authority across two entities, from monitors to procurement officers, and lower prices. I add communication between entities as a source of state performance. Where a task relies on it, digitizing it raised delivery with personnel, incentives, and authority unchanged.

The organizational economics of hierarchies studies how communication shapes the allocation of tasks and authority within organizations (e.g., Bolton & Dewatripont, 1994; Garicano, 2000; Dessein, 2002; Alonso *et al.*, 2008). In these models, cheaper communication changes who decides and what each level does. Most of the empirical work is set in private firms (McElheran, 2011; Giroud, 2013; Bloom *et al.*, 2014; Gumpert *et al.*,

[14]Before adoption, an indirectly managed center made 34.7 offers and enrolled 31.6 children, and the acceptance rate, averaged across centers, was 91.3%. Adding 5.30 children to the same offers would require a rate of 106%. The direct estimate of the change in the acceptance rate is 0.77 points (s.e. 1.20). Written queries are the family-to-agency channel I observe. The in-person channel, which might also operate, is not recorded.

2022), and extensions to the state have remained theoretical (e.g., Snowberg & Ting, 2019). The nearest empirical precedent, Mastrorocco & Teso (2023), uses the spread of the railroad and telegraph and finds that where communication improved, decision-making power was delegated. I bring the framework to the state with transaction-level data on who communicates with whom, and when. In this setting the law fixes who decides, so cheaper communication cannot reallocate authority. Instead, it moves what the state delivers.

Research on technology in the state has mostly studied technologies that change the information available to an entity. At the front line, they monitor delivery (Muralidharan *et al.*, 2016; Dal Bó *et al.*, 2021; Muralidharan *et al.*, 2021). In the administrative core the evidence is mixed: digitizing transfers between tiers of government cut leakage but slowed payments (Banerjee *et al.*, 2020), while giving officials program information cut processing time and raised output (Dodge *et al.*, 2023). I study the communication side of ICT through a platform built so that the entities of the state share information and coordinate. It raised what the state delivered for a given level of capacity.

Section 2 describes the setting, the reform, and the data, and Section 3 the research design. Section 4.1 shows how adoption changed communication and enrollment, Section 4.2 examines threats to identification and alternative explanations, and Section 4.3 shows who benefits and what the additional children cost. Section 5 concludes.

2 Institutional setting, reform and data

2.1 Childcare provision in Chile

The *Junta Nacional de Jardines Infantiles* (JUNJI, henceforth *the agency*) is Chile's largest public provider of early-childhood education, covering 53% of national preschool enrollment (Arias *et al.*, 2024). It runs 2,480 centers under 16 regional directorates and a national directorate. The sample covers traditional childcare centers serving children aged 0 to 4, excluding pre-kindergarten.[15] The agency directly manages 790 centers. The other 1,690 are *indirectly managed* by municipal governments, local authorities, or nonprofits with agency funding. For these centers, the agency sets standards, transfers funds per child attending, and supervises compliance, but does not manage daily operations.[16] Providers outside the agency were not reached by the reform.

The placement process is annual.[17] Families apply to one center during the enrollment window, which runs from October, when applications open, to March, when the school year starts. The national directorate sets the calendar and priority rules for every directorate, and the regional directorate checks eligibility and ranks applicants at each center through the enrollment system, once in the window. It also coordinates the process with its centers through documents, and each center offers a specific seat to each selected child, following the order of the ranking, and cannot alter it. A seat stays attached to a selected child until the outcome is recorded. In the enrollment system, closing the record frees the seat, and the center offers it to the next child on the list. Centers that do not use the system send documents to the regional directorate, which records the outcome for them. At the close of the cycle, each application stands as enrolled, declined, assigned but without family response (*no answer*), or waitlisted. Outside the window, seats open gradually, as children withdraw or transfer, which centers record or report to the directorate. The directorate re-ranks the waiting list at a higher frequency

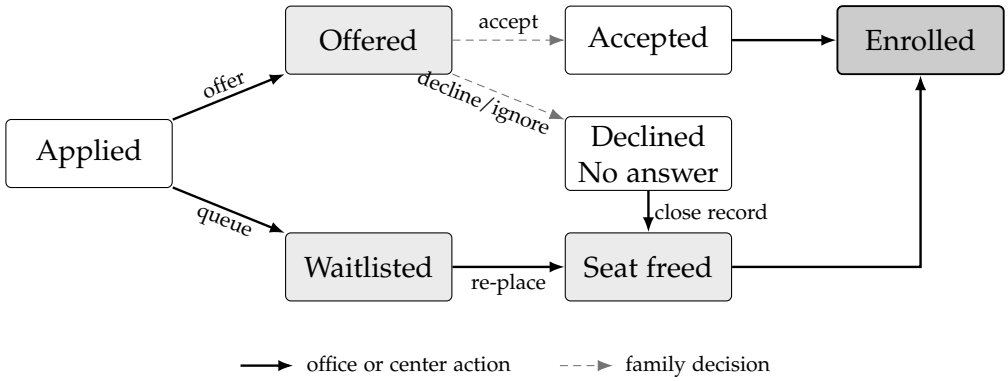
[15] Appendix G describes how the sample is constructed.

[16] During the pandemic, under article 14 bis of Decree 67/2010, the transfer was calculated on the average attendance of 2019 from 2020 to 2022 (JUNJI, 2022c, p. 5).

[17] The admission rules are set in JUNJI (2022b), the enrollment process in JUNJI (2021b), the functions of each department in the organizational resolution JUNJI, 2021a, art. 90, and the transfer program in its manual (JUNJI, 2019).

with new and remaining applications, including families that move their application to another center. Figure I traces one cycle.

FIGURE I
THE ENROLLMENT PIPELINE IN ONE ADMISSION CYCLE



Notes. This figure describes the enrollment pipeline of one center in one admission cycle. Each applicant is either offered a seat or placed on the waiting list. A family that receives an offer accepts, declines or does not answer. A seat attached to a declined or unanswered offer stays blocked until the center records that outcome. Once recorded, the center offers the seat to the next child on the waiting list. Solid arrows are steps taken by the agency or the center: the regional directorate ranks applicants through the enrollment system by the agency's priority rules, and the center offers the seats and records each outcome (the regional directorate records it for centers outside the enrollment system). Dashed arrows are family decisions. By construction, applications equal offers plus the waiting list, and offers equal enrolled plus declined plus no answer. The diagram follows the agency's admission rules (JUNJI, 2022b).

2.2 Official communications

Under Chilean administrative law, government actions must be documented and communicated to be legally valid. The standard instrument is the *oficio*, used for formal exchanges within and between public entities.[18] To produce one, a department requests a document number (*folio*), which the records office assigns to each official document. It then drafts the document, collects the required endorsements, obtains the final signature, and dispatches it with the number stamped on it (see Figure A.1, Panel A). Processing time is the internal interval from the request for the number to dispatch, measured in working days.

Directorates communicate with all their centers through documents on management, general guidelines, information and the coordination of services, such as vaccination

[18]Counterparts include other regional directorates, the national directorate, municipalities, ministries, regulators, and the centers themselves.

campaigns. Day-to-day operation differs by management type. Directly managed centers are run internally by the agency. Indirectly managed centers receive agency funds and send the directorate monthly reports and accounts as documents, including children who withdraw or move to another center (JUNJI, 2019, p. 32).

For enrollment, the regional directorate uses the enrollment system and communicates with its centers through documents. All centers receive as documents the rules and calendar of each enrollment window and, during it, follow-ups that monitor progress and ask centers to correct records that look inconsistent. Directly managed centers keep their enrollment records only in the agency's enrollment system (JUNJI, 2021b, § 2), and so do indirectly managed centers that use it. Centers outside the system send their enrollment forms to the directorate as documents, and the directorate enters them into the system and sends the results back the same way (JUNJI, 2022a). I do not observe which indirectly managed centers are outside the system, nor how many, so the indirectly managed group pools centers with and without it.

2.3 The reform and its rollout

Before the reform, the *oficio* workflow ran on paper (see Figure A.1, Panel A). An internal department drafted a paper document, collected approvals by hand, and sent it by courier. The process involved many handoffs and often long distances. On average, a document took 1.13 working days to leave the office (see Appendix Figure A.4).[19] Document handling absorbed between 30 and 50 percent of the time of any administrative procedure (Secretaría de Gobierno Digital, Gobierno de Chile, 2022).

The 2019 Digital Transformation Law introduced *DocDigital*, a central platform that carries the same procedure electronically (see Figure A.1, Panel B).[20] The law assigns entities to adoption groups, and the Digital Government Division schedules training and installation according to its own deployment capacity. Entities adopted one by one

[19]The sender department routed the document through its *oficina de partes*, the records office that receives, stamps and dispatches official correspondence. The receiving entity's records office logged the document on arrival and passed it to the responsible department.

[20]Law No. 21.180 mandates the platform as the only change, and leaves the steps, endorsements, signatures and content of a document as they were (Ministerio Secretaría General de la Presidencia, 2019, art. 1).

between 2020 and 2023 (see Table A.1 for the agency's directorates).[21] The platform does not require both sides to adopt. An entity that has adopted dispatches its documents digitally whether or not the receiver has the technology. If the receiver does not have it, the document arrives by email.

The reform directly changes the time and cost of producing a document, which is what the technology was designed to do. It changes the medium through which the agency's documents reach its receivers, not who decides, what a document is, or the steps that produce it.

Digitization should therefore improve the process at all of the agency's centers, directly and indirectly managed, since each relies on documents. Indirectly managed centers rely on them at more steps, and those outside the enrollment system send their enrollment forms and receive their results as documents. If digitization affects enrollment, the effect should appear at all agency centers and be larger at indirectly managed ones. Enrollment at providers outside the agency, which the reform did not reach, should not change.

2.4 Data

The main data source is the agency's internal registry of official communications, 2017 to 2023. The legal workflow ran on paper before the reform, but the agency logged the metadata of every document it sent in its own document management system: sender department, receiving entity, and timestamps. A sender department is a functional area of a directorate, such as finance, and a link is a pair of a sender department and a receiving entity (e.g., the finance department of a regional directorate and a municipality). After the reform, the agency kept the information in the same system and with the same fields, and it still stamps each document when it is sent.[22]

The estimation sample contains 146,170 documents sent by 119 departments across the 17 directorates (the 16 regional directorates and the national directorate) to 923 coun-

[21] The platform ran as version 2 throughout the rollout. Version 3, introduced in late 2023, requires the receiving entity to process documents through the platform as well, so the sample ends in 2023.

[22] DocDigital can be integrated with an entity's own document management system, which keeps recording its documents (Gobierno Digital, 2019, pp. 5, 8).

terparts, forming 2,842 links. I do not observe the text of a document, the communications the agency received, or transmission time after dispatch, nor, reliably, the management type of a receiving center. Communication outcomes are therefore measured on what the agency sends.

I classify each document by the formal role of the sender department toward its receiver, as fixed in the agency's organizational resolution (JUNJI, 2021a).[23] *Provision* is the correspondence of the departments charged with coordinating the enrollment process, with centers and with the entities the directorate must involve. It measures the agency's enrollment communication with all its centers. *Compliance* is the budget relationship, spending reported and budgets transferred, and goes only to the operators of indirectly managed centers. It measures the agency's link with those administration type centers. Three other classes serve as placebos.[24]

The Digital Government Division provided the *DocDigital* platform logs, aggregated by sender, receiver and month. They give the adoption date of every public entity, including the agency's directorates, and Section 3.2 uses these dates to define treatment.[25]

Enrollment comes from the agency's own enrollment system (*Sistema de Ingreso y Matrícula*), the system on which families apply online and which records each application's date and its final status at the close of the enrollment cycle. The enrollment system runs in parallel to the communication technology throughout the period, and the reform does not change it. Directly managed centers use it, and so do indirectly managed centers except those without access. The technology therefore reached all centers through the directorate's general communication. For indirectly managed centers outside the system, it also reached the enrollment process itself, because the directorate sends them their results as documents. The directorate enters their forms into the enrollment system, so their records appear in the same source. The same records have offers, waiting lists,

[23] Department functions are those set out in the agency's organizational resolution before adoption (JUNJI, 2021a). Appendix G reproduces the relevant articles and reports the pre/post stability of the class shares.

[24] Education-other is sent by the same education and strategy departments as provision, to education bodies outside the enrollment chain. Internal is correspondence between departments of the same regional directorate. The residual holds infrastructure, legal and human-resources correspondence.

[25] Adoption is the date the Division completes onboarding training with the entity, after which it is formally enabled to issue communications through the platform.

capacity, quality certification, and the household ranking used to classify children by income, for all of the agency's centers; staffing is observed only for the centers it manages directly. For indirectly managed centers, I also observe the monthly transfer paid for each center and the expenditure return each center files, itemized by spending category, including personnel. Enrollment is annual, one cycle per year, while adoption varies by month. Applications and enrollment are flows of children over the cycle, and capacity is the stock of seats. For providers outside the agency, the placebo group, enrollment comes from the Ministry of Education.[26]

Table I describes the variables. Management separates directly from indirectly managed centers, the comparison at the core of the paper. The table reports enrollment and initial capacity for each type, and the application cycle (applications, offers, unanswered offers, declines and waiting lists) for all centers together. Geography groups the 16 regional directorates into five macro-regions.

[26]The Ministry's *Subsistema de Información de Matrícula* receives reports from individual centers. Data are published by its Research Unit.

Table I
Sample description

	Mean	p10	p50	p75	p90	p99	N
Panel A. Communications							
Processing time (days)	1.03	0.00	0.00	1.00	3.00	12.00	146,170
Documents (per link-week)	2.43	1.00	1.00	2.00	4.00	19.00	60,914
Documents per link-week, by class							
Internal	4.36	1.00	2.00	4.00	8.00	50.00	16,109
Provision	1.28	1.00	1.00	1.00	2.00	5.00	3,795
Education–other	1.43	1.00	1.00	1.00	2.00	6.00	3,592
Compliance	1.85	1.00	1.00	2.00	3.00	8.00	12,266
Other	1.78	1.00	1.00	2.00	3.00	12.00	25,138
Documents							146,170
Links (sender department × receiver)							2,842
Sending directorates							17
Receiving entities							923
Panel B. Centers							
Applications	55.8	12.0	41.0	74.0	117.0	231.0	17,725
Offers	42.3	11.0	33.0	56.0	86.0	152.0	17,725
Enrolled	37.4	10.0	29.0	50.0	75.0	135.0	17,725
No answer	1.5	0.0	0.0	2.0	4.0	14.0	17,725
Declined	3.4	0.0	1.0	4.0	10.0	28.0	17,725
Waiting list	13.4	0.0	3.0	15.0	37.0	128.0	17,725
Enrolled children							
Indirectly managed	31.2	9.0	25.0	41.0	60.0	115.0	12,479
Directly managed	52.1	15.0	46.0	71.0	97.0	159.0	5,246
Capacity, seats (initial)							
Indirectly managed	68.5	24.0	52.0	98.0	116.0	210.0	1,690
Directly managed	101.1	44.0	96.0	140.0	178.0	248.0	790
Center-years							17,725
Centers							2,480
Indirectly managed							1,690
Directly managed							790
Municipalities							319
Regions							16
Macrozones							5

Notes. This table describes the estimation samples. Columns report the mean, selected percentiles and the number of observations. Panel A uses the communications registry, 2017 to 2023. Processing time is in working days per document; documents per link-week are computed over link-weeks with at least one document sent, by class. Panel B uses center-years, 2016 to 2023, in the sample. Applications, offers, unanswered offers, declines and waiting lists count children over the enrollment cycle, for all centers. Enrollment is reported by management type, and initial capacity is the seats in a center's first panel year, one value per center.

3 Research design

3.1 Identifying variation

The 2019 Digital Transformation Law (No. 21.180) mandated that public entities adopt the technology. Implementation was coordinated centrally by the *División de Gobierno Digital*, which operationalized the adoption schedule. The law places the agency's national directorate and its 16 regional directorates in the same adoption group, with the same requirements and deadline. Within that group, adoption dates differ because the Division enabled every public entity itself, providing and managing the platform and scheduling the training of each entity's staff (Gobierno Digital, 2019). I treat this variation in adoption timing as exogenous to potential outcomes. Section 4.2 tests this assumption.

The agency's directorates adopted between 2020 and 2021, and the public entities with which they communicate (municipalities, ministries, regulators, other public agencies) between 2020 and 2023 (see Figure II, Panel A).[27] The identifying variation differs by outcome. For communication, it is the date on which each sender–receiver link is first affected (see Figure II, Panel B). I compare the same link before and after adoption, with a single monthly treatment indicator for both communication outcomes. For enrollment, the identifying variation is the date on which each regional directorate adopted. When a regional directorate adopts, all of its centers, directly and indirectly managed, are treated at the same time.

3.2 Empirical strategy

I study two outcomes. First, I examine whether communication between a sender and a receiver changes when the technology reaches their communication link. Second, I move to the regional directorate, which coordinates enrollment in its region, and examine whether enrollment at its centers changes when it adopts.

[27] Appendix Table A.1 lists each directorate's adoption date, and Appendix Figures A.2 and A.3 show the adoption status of each link.

Communication outcomes. Let i index the sender department of a directorate that the registry records as the sender of a document (e.g., finance, front-line education), $r(i)$ the directorate of department i, j a receiving entity, and t time. I estimate:

$$Y_{ijt} = \tau \, AdoptionTech_{r(i)jt} + \gamma_{ij} + \delta_t + \phi_{w(t)} + \lambda_{r(i)} \cdot t + \varepsilon_{ijt} \tag{1}$$

where Y_{ijt} is processing time on documents, and a count of documents once the data are collapsed to the sender–receiver pair by week: only the unit of observation changes, never the right-hand side.

The treatment $AdoptionTech_{r(i)jt}$ equals one from the month in which the sender or the receiver of link (i, j) adopts, whichever comes first. A link is a pair, and either side's adoption can change it. When the sender adopts, it produces its documents digitally. When the receiver adopts, the number of documents the sender sends to it may change. Treating a link from the first adoption therefore leaves a clean comparison group of links where neither side has adopted yet.[28]

The term γ_{ij} is a sender-department by receiver fixed effect (absorbing time-invariant features of each link), δ_t is a calendar-month fixed effect, $\phi_{w(t)}$ is a week-of-year fixed effect (absorbing seasonality in administrative work), and $\lambda_{r(i)} \cdot t$ is a directorate-specific linear trend. The directorate trend is there because adoption is scheduled directorate by directorate: it absorbs any smooth drift in a directorate's own paperwork that could line up with the date the Division scheduled it.

The coefficient τ is the average effect of moving the link from the fully paper-based regime to one in which digital exchange is feasible for the sender, the receiver, or both, comparing the same link before and after, and net of shocks common to every link in a period, of week-of-year seasonality in administrative work, and of any smooth drift specific to the sender's directorate.

Standard errors are clustered by sender–receiver pair, the level at which treatment

[28] For ease of exposition, I use the same treatment for processing time as for document counts. Processing time runs inside the sender, so only the sender's adoption can change it, and links treated through the receiver alone pull the pooled estimate toward zero. Appendix Table A.2 defines treatment by the sender's adoption (Panel A) and reports each adoption state (Panel B); the effect on links treated through the receiver alone is small and not distinguishable from zero.

is assigned, because the timing is the minimum of the two adoption dates. Links that share a sender also share that sender's adoption date, so p-values come from the wild-cluster bootstrap (Cameron *et al.*, 2008) with Webb weights (Webb, 2014) on the regional directorates.

Enrollment outcomes. Each center's enrollment process is coordinated by its regional directorate, which exchanges documents with it. For enrollment I estimate, at the childcare-center level,

$$Y_{ct} = \tau \, AdoptionTech_{r(c)t} + \gamma_c + \mu_{m(c)t} + \varepsilon_{ct} \tag{2}$$

where Y_{ct} is an outcome (e.g., enrollment, applications, capacity utilization) for center c in year t. The treatment $AdoptionTech_{r(c)t}$ takes the value one if the center's regional directorate $r(c)$ has adopted the technology by year t. A directorate that adopts during year t counts as treated from t, although only the months of t after its adoption are exposed.

The term γ_c is a center fixed effect, and $\mu_{m(c)t}$ is a macro-zone by year fixed effect, where $m(c)$ is the macro-zone of the center's region. Unlike equation (1), the time controls are macro-zone by year effects rather than directorate-specific trends, because here treatment varies only by directorate and year. Identification exploits the staggered timing of adoption across regional directorates.

The coefficient τ compares centers in the same macro-zone and year whose directorates adopted at different dates, under the assumption that, absent the reform, outcomes would have evolved similarly across them. Treatment is assigned at the regional directorate, so the cluster is at the same level.

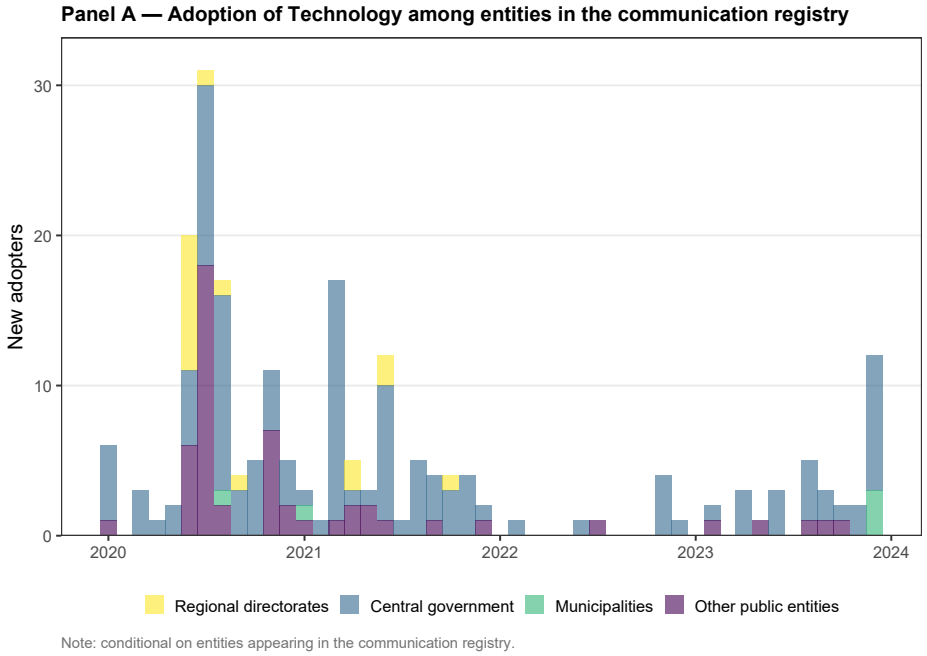
Staggered adoption. The two-way fixed effects can put negative weight on some treated observations when effects vary across cohorts or over time (Goodman-Bacon, 2021; de Chaisemartin & D'Haultfœuille, 2020). I therefore re-estimate both equations with the imputation estimator of Borusyak *et al.* (2024), which keeps their fixed effects and trends and imputes each treated outcome from observations not yet treated. Every directorate

had adopted by October 2021, so the imputation sample ends there for communication and, for enrollment, compares the 2020 cohort with the 2021 cohort in 2020. For enrollment I also report the estimator of Callaway & Sant'Anna (2021). Appendix Tables A.13 and A.14 report both sets of estimates.

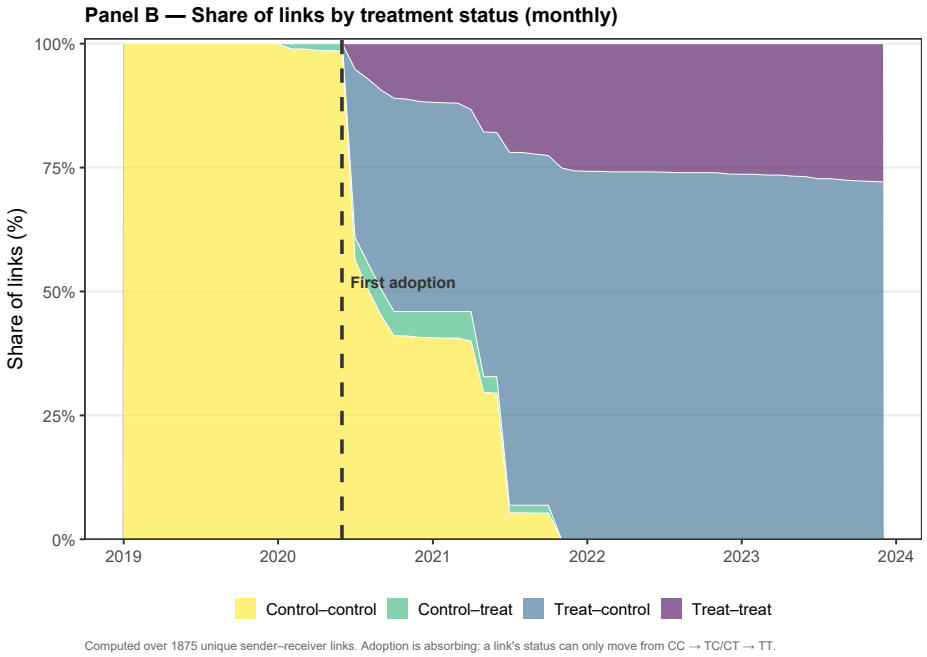
Threats to identification. Equations (1) and (2) assume that, absent the reform, outcomes would have evolved similarly across directorates that adopted at different dates. Three main concerns could break this assumption: early adopters were already on a different path, the pandemic drove the rollout, and regional shocks rather than the reform moved enrollment. I present the results first and address each concern, and the estimators for staggered adoption, in Section 4.2.

FIGURE II

ADOPTION OF THE TECHNOLOGY AND TREATED COMMUNICATION LINKS



Panel A: Adoption among entities in the communication registry.



Panel B: Share of treated communication links over time.

Notes. Panel A reports new adopters per month among entities present in the agency's communication registry: regional directorates of the agency, central government entities, municipalities, and other public entities, each entity counted in the month it activated the technology. Panel B reports the share of communication links in each treatment state as senders and receivers adopt: control–control (*CC*, neither the sender nor the receiver adopted), sender treated only (*TC*), receiver treated only (*CT*), and both treated (*TT*). Adoption is absorbing, so a link can only move from *CC* toward *TT*. The dashed vertical line marks the first adoption by one of the agency's directorates, June 2020. Shares are computed over the balanced panel of unique sender–receiver links observed in the registry.

4 Results

This section has three parts. Section 4.1 shows how adoption changed the agency's communication and enrollment at its centers, more at indirectly managed centers, with providers outside the agency as a placebo. Section 4.2 examines threats to identification and alternative explanations. Section 4.3 asks who benefits and what it cost.

4.1 Communication and enrollment

This subsection shows that adoption changed the agency's communication and increased enrollment at its centers.

Communication. I begin by studying the effect of the technology on the agency's communication. I look at three outcomes: processing time, the total number of documents, and the documents sent by the function of the sender department. The registry records every document sent by a department of the agency's directorates and its receiver: another department, another directorate, a center or its operator, or other public entities. The unit of analysis is the sender department–receiver pair.

Adoption makes document production faster. Processing time is the working days from the request for a document number to its dispatch. Figure III, Panel A, shows that processing time is flat before adoption and drops when the sender or the receiver adopts. The pooled estimate is a cut of 0.79 days on a pre-adoption mean of 1.13 working days, a 70% reduction, from next-day to same-day dispatch.[29] Because processing time excludes transit, this is a lower bound on the time the technology saves. For example, regional directorates routinely send documents to the national directorate, on average 772 km away and up to 2,193 km.[30] On paper, each one had to travel that distance, while a digital one arrives when it is sent.

Next, I turn to the total number of documents and find that it does not move. The

[29]The decline holds by receiver, from 55% to 86% of the pre-mean (Table II, Panel A), and by sender department, from 24% to 89% (Appendix Table A.3). The sign holds in every split, and two of the estimates are imprecise. Appendix Figure A.4 reports the distribution of processing time before and after adoption. The share of documents dispatched the same day rises, and the share that takes two days or more falls.

[30]Straight-line distances from each regional directorate to the national directorate in Santiago.

outcome is the number of documents each sender–receiver pair sends per week, in weeks in which it sends at least one. The pooled estimate is a fall of 0.24 documents on a pre-adoption mean of 2.47, a 9.9% decline that is not distinguishable from zero (Table II, Panel B, column 1). The confidence interval rules out increases larger than 7% of the pre-adoption mean.[31]

However, the composition of the flow changes. To study different tasks, I classify each document according to the function of the sender department toward its receiver. These functions did not change over the sample period, so the classification does not depend on adoption or on enrollment (see Section 2.2 and Appendix G). I classify as *provision* documents those sent by the departments responsible for coordinating enrollment. The *compliance* category consists of budget and reporting documents, which only indirectly managed centers receive. The flow internal to each regional directorate and the remaining categories show no change.

Provision documents rise in the months when the agency assigns seats. Panel B of Figure III shows that provision is flat before adoption and rises after it. The pooled estimate is an increase of 0.39 documents per active pair-week on a pre-adoption mean of 1.17, a 34% increase (see Table II, Panel B).[32] Inside the October-to-March window, when the agency assigns seats for the next academic year, provision rises 71% and does not move outside it (-1%; Table II, Panel C). Other education documents, sent by the same departments to entities outside the enrollment process, do not move inside the window. Outside it, they rise by 0.39 on a mean of 1.51 (26%), not distinguishable from zero. These patterns suggest that adoption increased the communication of the departments that coordinate enrollment in the enrollment window, and not other communication from the same departments.

Compliance documents, which only indirectly managed centers receive, drop in weekly number but spread over more weeks after adoption. Compliance falls 0.63 doc-

[31] Counts per pair could stay flat if each department started writing to more counterparts. This does not happen. At the directorate level, total documents sent and the number of counterparts each directorate writes to are flat after adoption (Appendix Table A.4).

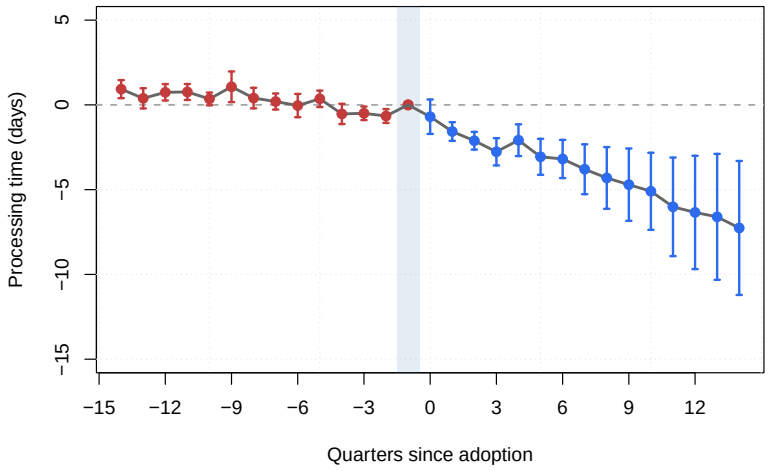
[32] I check the extensive margin by counting weeks with no documents, and provision also rises. Pairs send provision documents in more weeks, and they send more documents beyond the first (Appendix Table A.5).

uments per active pair-week on a pre-adoption mean of 1.71, a 37% decline (see Table II, Panel B). Counting weeks with no documents, pairs send these documents in 25% more weeks and send 58% fewer beyond the first in each week, so the total does not change detectably (Appendix Table A.5). This is the channel those centers share with the agency. Its change shows that the technology reached it.

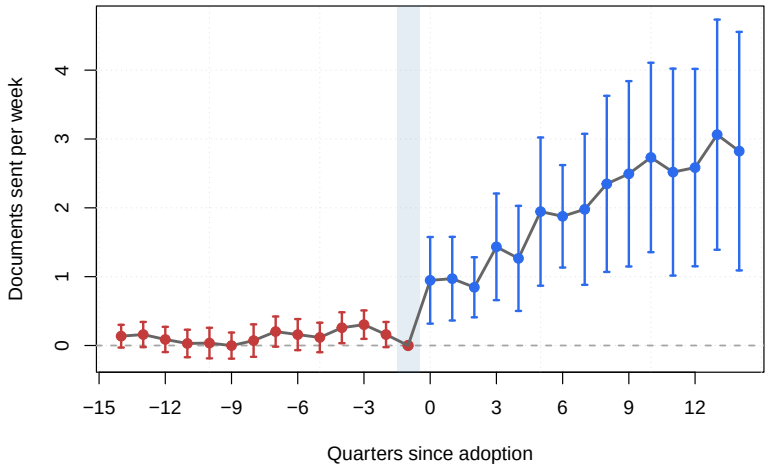
Adoption changed the speed and composition of the agency's communication. The departments responsible for coordinating enrollment send more documents in the months when the agency assigns seats. The technology therefore reached two channels: enrollment communication with all centers, and the channel shared only with indirectly managed centers. I next ask whether enrollment rose.

FIGURE III
EVENT STUDIES OF COMMUNICATION AROUND ADOPTION

A. Processing time



B. Provision documents



Notes. This figure plots event-study estimates of platform adoption on communication outcomes, at the sender department–receiver link level in event time. Panel A is processing time, the working days from the request for a document number to its dispatch. Panel B is documents sent per week by the departments responsible for coordinating enrollment. Both panels include link and month fixed effects, week-of-year dummies and directorate-specific linear trends. Standard errors are clustered by link, and p-values come from a wild-cluster bootstrap by regional directorate. Each dot is a coefficient relative to quarters -1 and -15, which are omitted; vertical lines are 95% confidence intervals. Panel B is estimated on external links only.

TABLE II

COMMUNICATION AFTER ADOPTION: PROCESSING TIME AND VOLUME OF DOCUMENTS

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Processing time (days), by receiver						
	All	Within regional directorate	Within the agency	To centers	To other public bodies	
$AdoptionTech_{st}$	-0.79***	-0.64	-0.64***	-0.97***	-0.87***	
	(0.24)	(0.69)	(0.18)	(0.06)	(0.15)	
Mean pre Y	1.13	1.16	1.16	1.13	1.05	
Effect, % of pre mean	-69.4	-55.5	-55.2	-85.5	-82.6	
Wild-boot p	0.070	0.648	0.043	0.016	0.026	
Dyads	2842	149	534	1454	783	
Observations	146170	69426	15061	36523	25147	
Panel B: Documents sent per week, by task class						
	Total	Internal	Provision	Education–other	Compliance	Other
$AdoptionTech_{st}$	-0.24	0.02	0.39***	0.08	-0.63***	0.10
	(0.21)	(0.57)	(0.12)	(0.17)	(0.06)	(0.31)
Mean pre Y	2.47	4.85	1.16	1.56	1.71	1.81
Effect, % of pre mean	-9.9	0.5	34.1	5.4	-37.1	5.5
Wild-boot p	0.335	0.973	0.032	0.544	0.049	0.719
Dyads	2771	146	563	357	344	1437
Observations	60914	16109	3795	3592	12266	25138
Panel C: Documents sent per week, by enrollment window (Oct–Mar vs. rest)						
	Provision		Education–other			
Enrollment window:	In	Off	In	Off		
$AdoptionTech_{st}$	0.84***	-0.01	-0.08	0.39		
	(0.20)	(0.14)	(0.31)	(0.25)		
Mean pre Y	1.18	1.14	1.60	1.51		
Effect, % of pre mean	71.0	-1.1	-5.2	25.9		
Wild-boot p	0.037	0.856	0.681	0.205		
Dyads	374	390	263	247		
Observations	1762	1801	1821	1635		
Dyad effects	Yes	Yes	Yes	Yes	Yes	Yes
Month effects	Yes	Yes	Yes	Yes	Yes	Yes
Week-of-year effects	Yes	Yes	Yes	Yes	Yes	Yes
Directorate linear trends	Yes	Yes	Yes	Yes	Yes	Yes

Notes. This table reports estimates of platform adoption on communication. In Panel A, the processing time measures the working days from the request for a document number to its dispatch, by receiver type. Each observation is a document. In both Panels B and C, the outcome are documents sent per week, and observations are sender department–receiver link-weeks with at least one document. Each column is a separate regression on its own sample. Panel B splits documents by the function of the sender department, as defined in the agency's organizational resolution. Provision documents are those sent by the departments responsible for coordinating enrollment. Compliance documents are the budget and reporting documents, which only indirectly managed centers receive. Education–other documents are sent by the same departments as provision to education entities outside the enrollment process. Internal documents stay within a regional directorate, and other documents are the remaining classes. Panel C splits provision and education–other documents by the enrollment window, October to March, against the rest of the year. The pre-adoption mean is taken before either side of the pair adopts. Standard errors, clustered by link, in parentheses; p-values from a wild-cluster bootstrap by regional directorate.

Enrollment at the agency's centers. Enrollment rises at the agency's centers and does not move at providers the technology did not reach. Figure IV shows that enrollment at the agency's centers is flat before adoption and rises after it. The pooled estimate is an increase of 4.17 children per center×year, 11% of the pre-adoption mean of 38.9 (see Table III). Providers outside the agency do not move (−0.35).[33]

I next check two other channels through which enrollment could rise. First, the agency could have expanded supply, opening centers or adding seats, but the number of centers (−0.14) and the seats per center (−0.94, negative if anything) do not move (see Table III). Second, adoption could have raised demand, with more families applying for a seat, but applications per 1,000 eligible children show no detectable change (−0.21, s.e. 3.39, on a base of 102).[34] Enrollment therefore increased for a given level of seats, and not because adoption made families more prone to apply.

The increase is larger at indirectly managed centers. Directorates digitized the general communication they send to both types of centers (Section 2.2). Indirectly managed centers also exchange more of their work with the directorate as documents, including, for those outside the enrollment system, their enrollment forms and results. If the effect on enrollment runs through these documents, it should be larger at these centers.

Table IV matches this prediction. At indirectly managed centers, enrollment rises by 5.30 children per center-year, 17% of the pre-adoption mean of 31.6. At directly managed centers the estimate is 0.83 and is not distinguishable from zero. In a regression that pools both types, the difference between them is 8.44 children.[35] It holds when the effect is allowed to vary with initial capacity (the capacity interaction is −0.23 per 10 seats, not distinguishable from zero), and with municipality-by-year effects, which compare

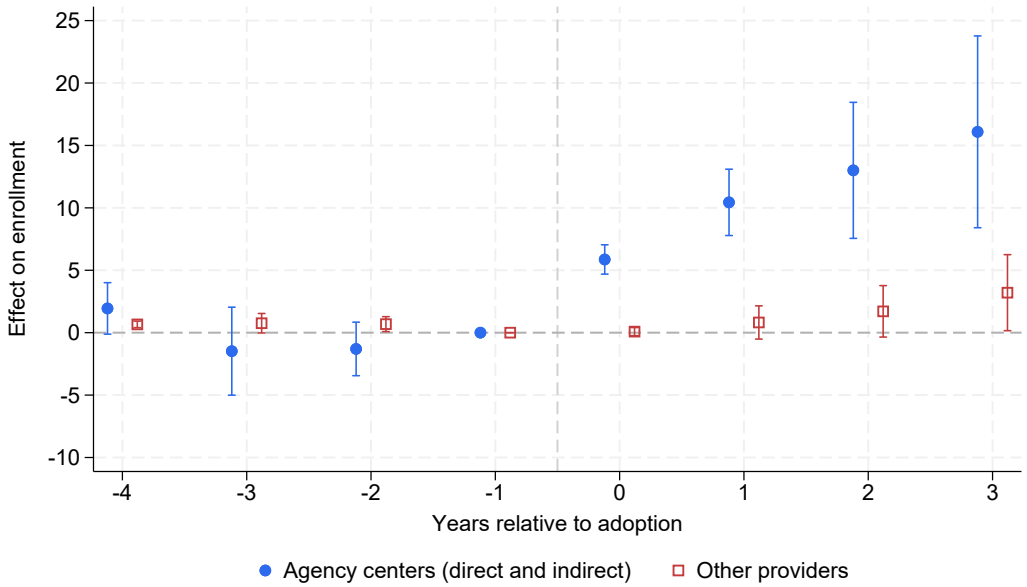
[33] Providers outside the agency are NGO *Integra* and private centers in the same municipalities, assigned the adoption date of their regional directorate. They share the municipality's demand shocks across time. They target different children, so this is a placebo test rather than a counterfactual.

[34] Applications per center do rise, by 4.17 (s.e. 0.73) on a pre-adoption mean of 56.1, 7% (Appendix Table A.6, column 1). Weighted by baseline center size, the estimate is −1.26 (s.e. 1.58). The three estimates are consistent with application records moving toward the smaller centers of the agency, with no more children applying in the municipality.

[35] In the pooled regression, the estimate for directly managed centers is −1.60. The difference is 7.64 when the effect also varies with initial capacity, and 7.73 with municipality-by-year effects (columns 4 and 5).

FIGURE IV

ENROLLMENT AROUND ADOPTION, THE AGENCY AND OTHER PROVIDERS



Notes. This figure plots event-study estimates of enrollment around the regional directorate's adoption of the platform. Each dot is a coefficient in years relative to adoption, with the year before adoption omitted; vertical lines are 95% confidence intervals. Blue circles are the agency's centers, which the reform reaches. Red squares are NGO and private providers pooled, which it does not: they are assigned the regional directorate's adoption year as a placebo. The two series are offset horizontally so their intervals do not overlap. Event time is binned at $+3$. Both specifications include center, year and macrozone$\times$year fixed effects; standard errors are clustered by region.

TABLE III
ENROLLMENT, SUPPLY AND DEMAND

Dependent Variable:	(1)	(2)	(3)	(4)	(5)
	Enrollment per center		Centers per municipality	Capacity, seats per center	Apps per 1,000 eligible
	The agency	Providers outside the agency			
$AdoptionTech_{rt}$	4.17***	-0.35	-0.14	-0.94	-0.21
	(0.42)	(0.32)	(0.10)	(0.65)	(3.39)
Wild-boot p	0.014	0.547	0.143	0.083	0.976
Mean Y (pre)	38.9	48.4	7.4	81.5	102.1
Observations	17725	24739	2463	15814	2423
Center effects	Yes	Yes	No	Yes	No
Muni effects	No	No	Yes	No	Yes
Year effects	Yes	Yes	Yes	Yes	Yes
Macrozone × year effects	Yes	Yes	Yes	Yes	Yes
Unit	center-year	center-year	muni-year	center-year	muni-year

Notes. This table reports difference-in-differences estimates of platform adoption by the regional directorate. Each column is a separate regression. Providers outside the agency are NGO and private centers, which the reform does not reach; they are assigned their region's adoption year. Standard errors, clustered by region (16), in parentheses; p-values from a wild-cluster bootstrap (Webb).

centers of both types in the same municipality and year.

The additional children come from more offers. Table V follows each year's applications through the admission process (see Figure I). At indirectly managed centers, offers rise by 5.54 per center-year and enrollment by 5.30. The acceptance rate does not change statistically, 0.77 points on a pre-adoption rate of 91.3%. At these centers, enrollment rises by almost as much as offers. Among the offers that do not end in enrollment, declines rise by 0.99 and offers with no answer fall by 0.75. The waiting list shortens by 0.95, imprecisely.[36] At directly managed centers, offers and enrollment do not move. Offers with no answer fall there too (-1.68 on a pre-adoption mean of 3.9), so that decline is common to both types. The rise in offers and declines points to a change in the admission process at indirectly managed centers.

[36] Appendix Table A.7 splits the pipeline by application date. Inside the window, offers rise by 3.18 with no change in applications and the waiting list shortens by 4.14. Outside it, applications, offers and enrollment rise together.

TABLE IV
ENROLLMENT BY MANAGEMENT TYPE

Dependent Variable:	(1)	(2)	(3)	(4)	(5)
	Enrollment per center				
	Indirect	Direct	Pooled	(3) + capacity	(4) + muni × year
$AdoptionTech_{rt}$	5.30***	0.83	-1.60**	-0.93	–
	(1.04)	(2.19)	(0.72)	(0.56)	
× Indirect			8.44***	7.64***	7.73***
			(1.11)	(0.93)	(0.97)
× Capacity (10 seats)				-0.23	-0.23
				(0.16)	(0.21)
Wild-boot p	0.028	0.817	0.000	0.000	0.000
Mean Y (pre)	31.6	57.5	38.9	38.9	39.2
Observations	12479	5246	17725	17725	17403
Center effects	Yes	Yes	Yes	Yes	Yes
Year effects	Yes	Yes	Yes	Yes	Yes
Macrozone × year effects	Yes	Yes	Yes	Yes	Yes
Muni × year effects	No	No	No	No	Yes

Notes. This table reports difference-in-differences estimates of platform adoption on enrollment per center, by management type. Columns (1) and (2) are separate regressions for indirectly and directly managed centers. Columns (3) to (5) pool all centers and add the interaction of adoption with an indicator for indirect management; there the adoption row is the effect in directly managed centers under time effects common to both types. Column (4) adds the interaction with initial capacity, demeaned, in tens of seats. Column (5) adds municipality×year effects to column (4); they absorb adoption. The wild-bootstrap p-value refers to adoption in columns (1) and (2) and to the interaction with indirect management in columns (3) to (5). Standard errors, clustered by region (16), in parentheses; p-values from a wild-cluster bootstrap (Webb). Sample: 2016–2023.

TABLE V
THE ENROLLMENT PIPELINE, BY MANAGEMENT TYPE

	(1) Offers	(2) Acceptance rate	(3) Declined	(4) No answer	(5) Enrolled	(6) Waiting list
Panel A: Indirectly managed centers						
$AdoptionTech_{rt}$	5.54***	0.77	0.99***	-0.75***	5.30***	-0.95**
	(1.19)	(1.20)	(0.26)	(0.20)	(1.04)	(0.36)
Wild-boot p	0.029	0.731	0.055	0.092	0.028	0.162
Mean Y (pre)	34.7	91.3	1.4	1.6	31.6	10.6
Observations	12479	12473	12479	12479	12479	12479
Panel B: Directly managed centers						
$AdoptionTech_{rt}$	-0.53	0.78	0.32	-1.68***	0.83	1.93*
	(2.29)	(1.16)	(0.32)	(0.36)	(2.19)	(0.97)
Wild-boot p	0.853	0.586	0.390	0.062	0.817	0.163
Mean Y (pre)	64.7	88.7	3.3	3.9	57.5	18.3
Observations	5246	5233	5246	5246	5246	5246
Center effects	Yes	Yes	Yes	Yes	Yes	Yes
Year effects	Yes	Yes	Yes	Yes	Yes	Yes
Macrozone $\times$ year effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes. This table reports difference-in-differences estimates of platform adoption on the enrollment pipeline, for all applications of the year, inside and outside the enrollment window. Each cell is a separate regression at the center-year level. By construction, applications equal offers plus the waiting list, and offers equal enrolled plus declined plus no answer. The acceptance rate is enrolled over offers, in percent, for center-years with at least one offer. No answer is an offer the family did not answer; declined is a recorded refusal. Standard errors, clustered by region (16), in parentheses; p-values from a wild-cluster bootstrap (Webb). Sample: 2016–2023.

This pattern is consistent with a document-dependent handoff. In each window, the regional directorate ranks the applicants to each center once, and the center then offers its seats in that order. A seat is freed when the outcome of its offer is recorded in the enrollment system, and the center offers it to the next child on the list (JUNJI, 2021b). Directly managed centers record each outcome in the system themselves. Indirectly managed centers report withdrawals and transfers to the directorate as documents. Those outside the system also receive their results from the directorate as a document and return their records the same way, on paper before the reform (JUNJI, 2021b, 2022a).[37] The reform digitized this exchange and left the enrollment system unchanged.

The results line up with this reading. The directorate's enrollment communication rises with adoption (Table II), and enrollment rises more at indirectly managed centers, where more of the process runs through it (Table IV). Offers rise and the acceptance rate does not (Table V), so the additional children come from more offers, the margin the center moves, not from families accepting more of them.[38] I do not observe which step changed or which centers are outside the enrollment system, so I present this as a plausible mechanism. Section 4.2 discusses alternatives, including differences between management types in staff and payment.

Adoption raised enrollment at the agency's centers and left providers outside the agency unchanged. Centers, capacity, and applications per eligible child do not move. The agency therefore enrolled more children for a given level of seats. The increase is larger at indirectly managed centers and comes through more offers, a pattern consistent with a document-dependent handoff. Section 4.2 examines threats to this result and alternative explanations.

[37] The 2023 guidelines set out the procedure for centers outside the enrollment system. The center sends its enrollment forms to the regional directorate on paper, the directorate enters them into the system, runs the prioritization and sends the results back as a document, and the center returns its enrollment records the same way (JUNJI, 2022a, pp. 9, 16, 20). The guidelines set the release of a seat at its record in the system and do not say whether a center outside the system waits for the directorate's record before offering it again.

[38] Processing time shows that the technology reached the documents.

4.2 Threats to identification and alternative explanations

The design assumes that, absent the reform, communication and enrollment would have evolved similarly across directorates that adopted at different dates. Below, I discuss three possible concerns: that early adopters were already on a different path, that the pandemic drove both the rollout and enrollment, and that regional shocks rather than the reform moved enrollment. I also check that the estimates do not depend on how two-way fixed effects weight the staggered adoption. I then consider four alternative explanations for the enrollment effect.

Early adopters were already on a different path. If directorates that adopted in 2020 were already sending more documents, or shifting them toward their centers, the communication estimates would pick up that change. To check this, I compare early and late adopters before adoption. For each directorate, I use the year-to-year change in total documents, in the shares of external and internal documents, and in provision and compliance documents. The differences are not distinguishable from zero (Appendix Figure A.5). These estimates are imprecise, especially for compliance, so the check rules out only large differences in trends.

The same concern applies to enrollment. If centers in early-adopting directorates were already enrolling more children, the enrollment estimates would pick up that trend. I run the same comparison on center-level applications, waiting-list records, vacancies, enrollment turnover, capacity, the worker-to-child ratio and the share of indirectly managed centers. All differences are small and not distinguishable from zero (Appendix Figure A.6). At the directorate level, enrollment, applications and waiting lists show the same pattern (Appendix Figure A.5).

The pandemic. The rollout ran during the pandemic period, so both effects could reflect pandemic shocks that differed across regions. If so, pandemic conditions should predict adoption dates. I use four monthly series by region: days under mandatory quarantine, emergency visits attributed to COVID for all ages, the same visits for ages

one to four, and the share of days at reopening phases.[39] I regress adoption of the technology on each measure over 2017 to 2021, with directorate and month fixed effects. None shows an association with adoption timing that is distinguishable from zero (see Appendix Table A.8). The point estimates range from -0.7 to 17.6 percentage points per standard deviation (wild p from .37 to .82), so the test rules out only large associations. Moreover, for enrollment, there is a second check. Providers outside the agency operate in the same municipalities and face the same quarantines, and their enrollment does not move (Appendix Table A.11).

A remaining concern is pandemic exposure. The early cohort spent fewer days under lockdown in March to May 2020 before adopting, 2.4 against 8.7, and most of this difference comes from the Metropolitan region (see Appendix Table A.9).[40] If the enrollment effect came from this difference, it should shrink once the difference is controlled for or removed. I test this over 2016 to 2020, the window in which the late cohort is a not-yet-treated comparison, by controlling for lockdown days and emergency visits in the enrollment specification (Appendix Table A.10). In this sample, the estimate does shrink, from 5.67 to 2.37 and 2.96, and loses statistical significance, with a standard error three times larger, a sign of lost precision. Without the Metropolitan region, the cohorts do not differ in exposure and the estimates are 5.52, 4.34 and 6.58 (wild p below .07), so the effect holds where the exposure difference is gone. Over the whole sample, interacting lockdown exposure with year indicators leaves the point estimate at 3.95 against 4.17 and raises its standard error from 0.42 to 1.67 (Appendix Table A.12, column 5). I read these checks as reasonably conclusive that the pandemic does not drive the whole effect, although it may account for part of it.

Regional shocks. Demographic change or regional policy could move outcomes in some regions and not others. The communication estimates control for this with directorate specific linear trends. For enrollment, I add several regional controls (see

[39] Quarantine days and reopening phases come from the Ministry of Health's *Paso a Paso* records, and emergency visits from its emergency-care records (DEIS), all aggregated to the region and month. Appendix G gives the definitions.

[40] The Metropolitan region had the longest lockdowns, 113.5 days in 2020 against at most 64.4 in any early-adopting directorate. Without it, the remaining fifteen directorates hold 74% of centers.

Appendix Table A.12). With year effects only, the estimate is 3.99 but imprecise. The baseline specification adds macro-zone by year effects and gives 4.17 (wild $p = .014$). Adding directorate-specific linear trends on top gives 4.04 (wild $p = .024$), and adding the child population of the municipality gives 4.18 (wild $p = .015$). The point estimate stays between 3.99 and 4.18.

Staggered adoption. Every directorate adopts within the sample, so there are no never-treated units. For enrollment, with two adoption cohorts, the effect is identified from the year in which only the first cohort has adopted and from the second cohort's switch relative to the first. With staggered adoption, two-way fixed effects use units already treated as controls for later adopters, and those comparisons receive negative weight, which biases the estimate when effects differ across cohorts or over time. They cover close to half of the treated link-weeks in the communication sample and a fifth of the treated center-years in the enrollment sample. The imputation estimator of Borusyak *et al.* (2024) avoids them by predicting each treated outcome from observations not yet treated.

I check both sets of results with the alternative estimators. The imputation estimator uses only the treated observations that still have a not-yet-treated comparison, 10 to 30% of treated link-weeks and 6.8% of treated center-years, so its estimates are for the period right after adoption. For communication it gives -0.88 days for processing time, 0.73 provision documents and -0.35 compliance documents per active pair-week, against -0.79, 0.39 and -0.63 under two-way fixed effects, with standard errors about twice as large (Appendix Table A.13). The signs hold and the magnitudes are of the same order. For enrollment it gives 6.15 children (standard error 1.05) against the 4.17 (0.42) of the main specification, and the estimator of Callaway & Sant'Anna (2021) gives 4.30 (3.34) (Appendix Table A.14). The signs hold here too, with larger standard errors, because only the 2020 cohort in its adoption year has a not-yet-treated comparison. These estimators confirm the direction and the order of magnitude of the effect but with less precision.

Alternative explanations. I consider four alternatives. First, digitization may have freed staff time at the center. I do not observe staff time, so I test its implication: the increase should be larger where administrative staff is scarce. Among directly managed centers, where I observe staff, the effect is larger without dedicated administrative staff, but the difference is not statistically significant and the estimate is imprecise. Among indirectly managed centers, the effect does not differ at the capacity threshold at which the agency funds an administrative assistant (see Table VI).[41]

Second, families may drive the increase. Families take three decisions that I observe: whether to apply, whether to write to the directorate to ask for a different center, and whether to accept an offer. The first two concern where a child applies. Applications per eligible child in the municipality do not change (Table III, column 5), but applications per indirectly managed center rise outside the window, by 5.61 on a pre-adoption mean of 9.1, and not inside it (Appendix Table A.7). This rise is more likely a move of applications toward these centers than more families applying. Outside the window, such a move starts with the family (JUNJI, 2019), but the written channel it uses to ask for a different center does not change (Appendix Table A.15).[42] In-person requests are not recorded, so families may drive part of this move. A higher acceptance rate could also raise enrollment, but it does not change (see Table V). With offers held at their pre-adoption level, the additional 5.30 children would require an acceptance rate of 106%.[43]

Third, children may have moved from directly to indirectly managed centers. If so, enrollment at directly managed centers would fall by about 11 children per center a year.[44] The estimate is 0.83, with a standard error of 2.19 (Table IV, column 2), and enrollment across the agency rises by 4.17 children per center (Table III). A fall of 11 children lies outside the 95% interval of the estimate, so the increase is not a transfer

[41] Columns 1 and 2 of Table VI are directly managed centers, where staff records show which centers have administrative staff. Columns 3 and 4 are indirectly managed centers, where staff is not recorded; there, the agency's transfer funds an administrative assistant only when the center has a capacity of 100 seats or more (JUNJI, 2019).

[42] Written queries are counted inside and outside the enrollment window. They are the family-to-agency channel I observe. The registry has no query type, so the count includes every written query a directorate receives.

[43] Before adoption, an indirectly managed center made 34.7 offers and enrolled 31.6 children.

[44] About 9,000 additional children a year (see Table VIII, Panel A) over roughly 790 directly managed centers, the difference between the 2,480 centers of Table VII, Panel A, and the 1,690 of Panel B.

between the two types.

Fourth, indirectly managed centers are paid per child attending, so an additional child brings revenue. A permanent incentive is absorbed by the center fixed effects, so to drive the effect it would have to be active at adoption. During the rollout the pandemic suspended payment by attendance, and transfers were computed on each center's average attendance in 2019, so an additional child brought no additional revenue (JUNJI, 2022c). If the effect ran through the payment, it should be absent in the years when a child brought no revenue. Over 2016–2020 the effect is 5.68, against 5.30 over the full sample (Appendix Table A.16, Panel B).[45]

The next subsection asks who the additional children are and what it costs to enroll them.

[45]Transfers tracked enrollment until 2019, with an elasticity of 0.29, and stopped tracking it in 2020 (Appendix Table A.16, Panel A). Before 2020 an additional child was worth about US$650 a year, 0.29 times the US$2,211 accepted expenditure per enrolled child (Table VIII, Panel B). The rule applied to all indirectly managed centers, so it cannot produce a difference between cohorts, but it does not settle what centers expected.

TABLE VI
ALTERNATIVE EXPLANATIONS: ADMINISTRATIVE STAFF

	(1)	(2)	(3)	(4)
Dependent Variable:	Enrollment per center			
	Directly managed		Indirectly managed	
$AdoptionTech_{rt}$	2.31	1.99	5.28***	5.22***
	(2.68)	(2.58)	(1.05)	(1.11)
× Any admin staff	-3.72	-1.25		
	(2.83)	(2.40)		
× Capacity (10 seats)		-0.32*	-0.05	
		(0.16)	(0.14)	
× 1[capacity ≥ 100]				0.27
				(1.40)
Wild-boot p	0.254	0.637	0.730	0.854
Mean Y (pre)	57.6	57.6	31.6	31.6
Observations	5048	5048	12479	12479
Center effects	Yes	Yes	Yes	Yes
Year effects	Yes	Yes	Yes	Yes
Macrozone × year effects	Yes	Yes	Yes	Yes

Notes. This table reports difference-in-differences estimates of platform adoption on enrollment per center, within each management type. Columns (1) and (2) are directly managed centers with staff records; columns (3) and (4) are indirectly managed centers. Any admin staff indicates that the center had administrative staff in 2018–19; it is observed only for directly managed centers. Capacity is initial capacity, demeaned, in tens of seats. The wild-bootstrap p-value refers to the interaction with admin staff in columns (1) and (2), with capacity in column (3), and with the 100-seat indicator in column (4). Standard errors, clustered by region (16), in parentheses; p-values from a wild-cluster bootstrap (Webb). The agency funds an administrative assistant in indirectly managed centers with a capacity of 100 seats or more (JUNJI, 2019).

4.3 Who benefits and what it cost

The enrollment effect raises two questions. Which households do the additional children come from, and how much does it cost to enroll them?

Most of the additional children come from the poorest 60% of households. At indirectly managed centers, 4.04 of the 5.38 additional children, 75%, come from this group (see Table VII, Panel B).[46] Across all centers the share is 74% (Panel A). Relative to the population, the increase is therefore progressive.[47] Relative to the agency's own enrollment, it keeps the same proportions. Before adoption the bottom 60% already held 67% of enrollment, and the share of the increase is not distinguishable from it. This is consistent with the reform leaving the targeting criteria unchanged.

What an additional child costs depends on whether staff scale with enrollment. Panel B of Table VIII shows the cost structure of indirectly managed centers in 2019. They spent US$2,211 per enrolled child a year. The inputs that scale with one more child, such as materials, utilities and cleaning, were US$158 of that.[48] The rest is mostly personnel spending. At these centers, adoption raised enrollment by 4.5 children, 14.6% of its pre-adoption mean (Appendix Table A.17, column 1). Two scenarios bound its cost. In the first, the additional children fill seats in groups that already have staff, and each costs the inputs that scale with a child. In the second, staff grow in proportion to enrollment, and each costs the running cost of a seat. The centers' personnel spending does not rise significantly, but the test is imprecise (Appendix Table A.17).[49]

Across the 1,690 indirectly managed centers, the effect means about 9,000 additional

[46] Panel B runs on 2017–2023, when the household ranking is complete. On this sample the effect is 5.38, against 5.30 on 2016–2023.

[47] For the 75% to reflect household composition alone, households below the cutoff would need twice as many children aged 0–4 per household as those above it.

[48] For 2019, the accepted-expenditure returns of the 1,736 transfer-financed centers, CLP 172,506,110,833 over 111,479 enrolled children. Personnel is wages, fees and severance. The child-scaling inputs are utilities, materials, cleaning and similar items. Food is provided separately and excluded. Pesos of 2019 at 700 CLP/USD.

[49] Log personnel spending moved by −1.1%, and its 95% interval rules out increases above 3.1% (column 6). Staff growing in proportion to enrollment would imply 13.6%. The same test in pesos is noisier and does not rule out that growth (column 2). Expenditure returns record only spending of the program transfer, so staff paid from a sponsor's own funds would not appear. In 2020–22 transfers were computed on each center's average attendance in 2019, not on current attendance, which flattens spending in those years. Staff are set per group by child-to-staff ratios, so an additional child requires hiring only when a group exceeds them (Ministerio de Educación, 2010, art. 10).

children a year (a 95% interval gives 3,600 to 16,700 children, see Table VIII, Panel A). Their cost is US$1.4 million a year in the first scenario and US$19.8 million in the second (Panel C). Neither includes construction.

The cost exercise bounds what the agency spends on the same children. It does not value the seats to the children or their families. I do not observe where these children were before they enrolled, at home with a parent or a relative, or in another center.[50] It also leaves out the time of the agency's staff, any cost or benefit to the family, and the cost of the platform.[51]

Overall, indirectly managed centers enrolled about 9,000 more children a year for a given level of seats, three in four from the poorest 60% of households. Depending on whether staff grew with them, these children cost US$1.4 to US$19.8 million a year.

[50] Enrollment does not fall at directly managed centers, 0.83 (s.e. 2.19), or at providers outside the agency, −0.35 (s.e. 0.32), see Table IV, column 2, and Table III, column 2. Both are estimates at the center level. They do not identify where any one child came from.

[51] The platform was budgeted at US$4.8 million, plus US$1.6 million a year in operation from 2022, for all the public entities that use it (Dirección de Presupuestos, 2018). It is a shared fixed cost, and I do not observe the share that corresponds to this agency.

TABLE VII
WHO BENEFITS: ENROLLMENT BY HOUSEHOLD RANKING

	(1) All	(2) Bottom 60	(3) Above 60
Panel A: All centers			
$AdoptionTech_{rt}$	4.20***	3.11***	1.09***
	(0.47)	(0.51)	(0.29)
Share of the gain (%)	100.0	74.0	26.0
Share of pre-adoption enrollment (%)	100.0	65.5	34.5
p: share of gain = pre-adoption share		0.432	
Wild-boot p	0.019	0.036	0.065
Mean Y (pre)	39.1	25.6	13.5
Observations	16041	16041	16041
Centers	2480	2480	2480
Panel B: Indirectly managed centers			
$AdoptionTech_{rt}$	5.38***	4.04***	1.34***
	(1.15)	(0.91)	(0.26)
Share of the gain (%)	100.0	75.1	24.9
Share of pre-adoption enrollment (%)	100.0	66.9	33.1
p: share of gain = pre-adoption share		0.145	
Wild-boot p	0.029	0.033	0.025
Mean Y (pre)	32.0	21.4	10.6
Observations	11216	11216	11216
Centers	1690	1690	1690
Center effects	Yes	Yes	Yes
Year effects	Yes	Yes	Yes
Macrozone × year effects	Yes	Yes	Yes

Notes. This table reports difference-in-differences estimates of platform adoption on enrollment per center, split by the household ranking (RSH). Bottom 60 is children from households in the poorest 60% of Chilean households, as classified by the RSH. This is the cutoff of the agency's priority model. Above 60 is children above it or with no bracket on record.

TABLE VIII
THE COST OF ENROLLING THE ADDITIONAL CHILDREN

		95% interval	
	Estimate	Lower	Upper
Panel A. Effect and scale, indirectly managed centers			
Additional children per center-year	5.30	2.15	9.90
Additional children per year ($\times$ 1,690 centers)	8,954	3,633	16,737
Panel B. Cost of a seat, 2019 (USD per child-year)			
Running cost of a seat (accepted expenditure)	2,211		
Inputs that scale with one child	158		
Exchange rate (CLP per USD)	700		
Panel C. Cost of enrolling the additional children, per year (USD million)			
Fixed staff	1.4	0.6	2.6
Staff in proportion to enrollment	19.8	8.0	37.0

Notes. This table prices the additional enrollment at indirectly managed centers. Panel A is the effect at those centers (Table IV, column 1), with its 95% wild-cluster-bootstrap interval (Webb), multiplied by the number of indirectly managed centers in the estimation sample; the interval treats that number as fixed. Panel B is the 2019 running cost of a transfer-financed seat (accepted operating expenditure per enrolled child) and the inputs that scale with one more child, from the centers' expenditure returns. Panel C multiplies Panel A, row 2, by each cost in Panel B: fixed staff uses the inputs that scale with a child, and staff in proportion to enrollment uses the running cost of a seat. Neither includes construction.

5 Conclusion

This paper asked whether better communication between the entities of the state helps it use the capacity it already has. I show how a platform that digitized official communication raised the enrollment of children in Chile's public childcare agency. Then, the paper follows the technology to a real outcome: the children enrolled. The platform changed how documents move and left who decides, what a document is and the steps that produce it untouched. Document processing time fell by 70% and documents from the departments that coordinate enrollment rose by 34%, concentrated in the enrollment window. Enrollment rose by 11% over its pre-adoption mean, and centers, capacity and applications per eligible child did not move. The increase is concentrated in indirectly managed centers, where enrollment relies more on documents. Three in four of the additional children came from the poorest 60% of households. The agency enrolled about 9,000 more children a year for a given level of seats, at a cost between US$1.4 million and US$19.8 million a year, depending on whether staff grew with them.

The evidence suggests that communication affects what the state delivers through the tasks that rely on it. Both types of centers received the same general communication, so communication in general did not raise enrollment on its own. Enrollment rose where part of the enrollment process traveled as documents, a pattern consistent with a document-dependent handoff, in which the directorate records each outcome and returns the results to these centers as documents. Who decides, what a document is and the steps of the procedure were fixed by law, and the reform changed no incentive. Communication alone moved output, through the task that relied on it.

Many public services are organized the same way. A coordinating office registers records that front-line units send on paper: hospital networks and their clinics, school districts and their schools, benefit agencies and their local offices. Where the office's decisions wait on those records, digitizing the channel should raise output. Where the units already report through a shared system, there is less for the channel to move.

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Appendix

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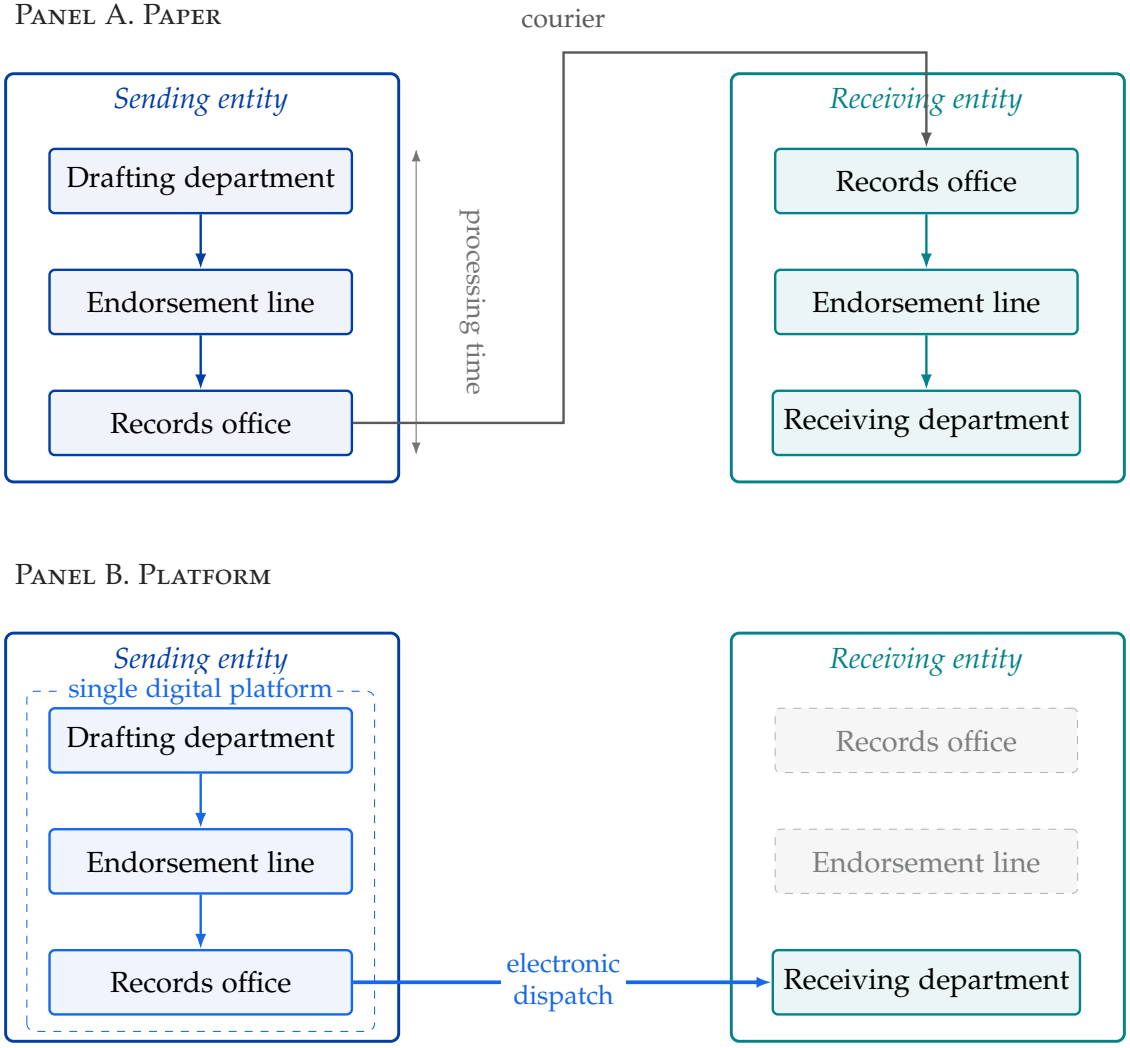
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G. Data and measurement. Sample of centers, applications, enrollment year, household income, communications, and classification of documents (Table A.18, Figure A.7)

A. The rollout

FIGURE A.1
THE OFFICIAL-COMMUNICATION WORKFLOW, ON PAPER AND ON THE PLATFORM



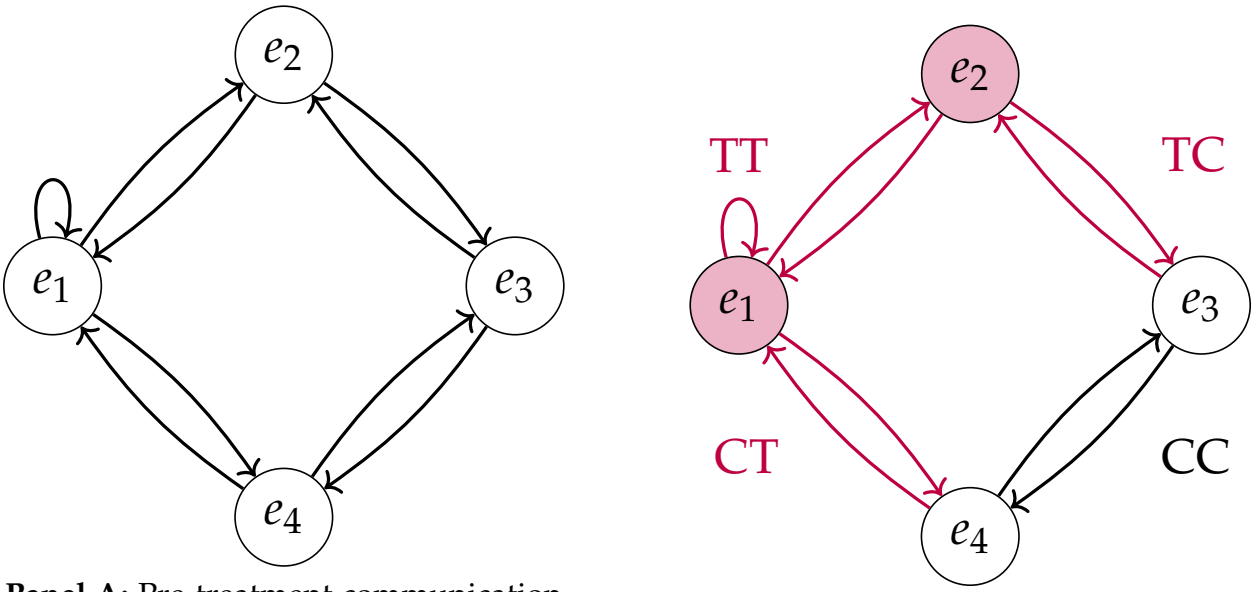
Notes. This figure shows how a document moves between two entities. Panel A shows the paper workflow and Panel B the platform. Boxes are departments and solid arrows are the steps a document must clear. Dashed boxes in Panel B are receiving-side steps that the platform leaves to the receiving entity. The bracket marks processing time, the working days from the request for a document number to dispatch. Transmission after dispatch is not observed.

Table A.1
Adoption dates of the agency's directorates

Date	Adopters	Date	Adopters
2 Jun 2020	1	17 Aug 2020	1
10 Jun 2020	2	10 Sep 2020	1
15 Jun 2020	1	19 Apr 2021	1
19 Jun 2020	1	28 Apr 2021	1
21 Jun 2020	1	8 Jun 2021	1
22 Jun 2020	3	24 Jun 2021	1
10 Jul 2020	1	12 Oct 2021	1
			Total directorates: 17

Notes. This table reports the date on which each of the agency's 17 directorates, the 16 regional directorates and the national directorate, adopted the platform. Adoption is absorbing. Excluding the national directorate, which oversees no centers, eleven regional directorates adopted in 2020 and five in 2021. These are the two cohorts of the enrollment analysis. Source: DocDigital administrative records.

Figure A.2
Adoption status of a sender–receiver link

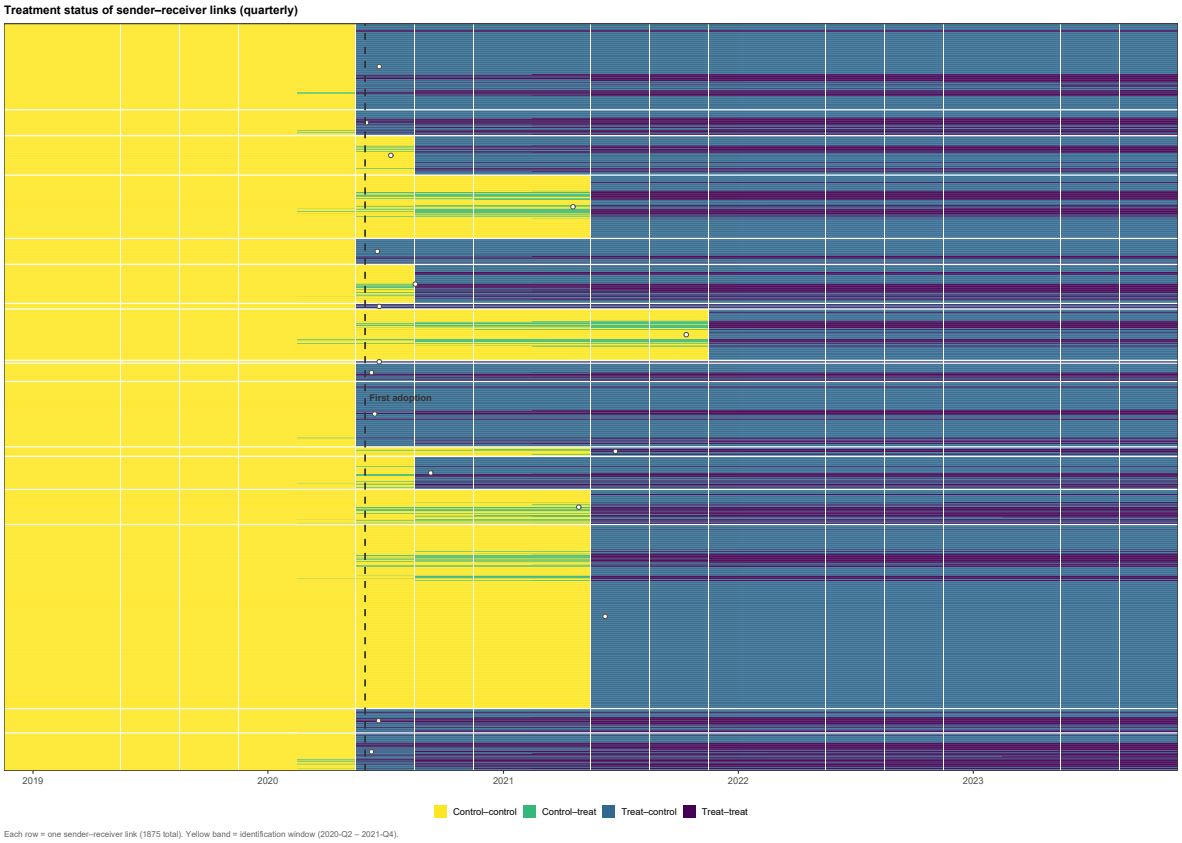


Panel A: Pre-treatment communication network.

Panel B: Post-treatment network.

Notes. This figure illustrates the four adoption states of a sender–receiver link. Nodes are public entities and edges are communication links. In *CC* neither the sender nor the receiver has adopted, in *TC* only the sender has, in *CT* only the receiver, and in *TT* both. A link is treated from the first adoption by either side, and treated links are shown in purple. The diagram is schematic.

Figure A.3
Adoption status of each link over time



Notes. This figure shows the adoption state of each sender–receiver link over time. Each row is a link and each column a quarter. Colors give the state of the link, *CC* (neither side has adopted), *CT* (only the receiver), *TC* (only the sender) and *TT* (both). The dashed line marks the first adoption, from which the link is treated.

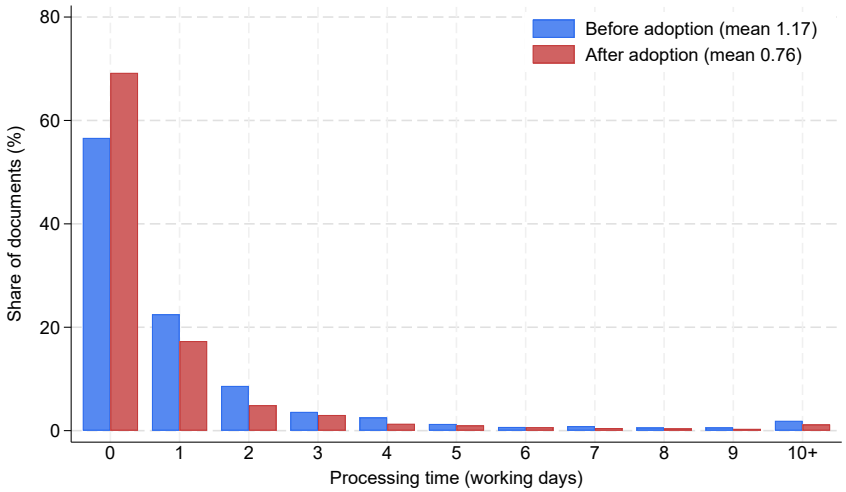
TABLE A.2
PROCESSING TIME BY ADOPTION STATE OF THE LINK

	(1) All	(2) Within regional directorate	(3) Within the agency	(4) To centers	(5) To other public bodies
Panel A: Sender adopted, with or without the receiver					
$Adoption_t$, sender adopted (TC or TT)	−0.79***	−0.64	−0.64***	−0.97***	−0.87***
	(0.24)	(0.69)	(0.18)	(0.06)	(0.15)
Mean pre Y	1.13	1.16	1.16	1.13	1.05
Wild-boot p, TC or TT	0.070	0.648	0.043	0.016	0.026
Share of observations, TC or TT	0.348	0.334	0.285	0.374	0.384
Dyads	2,842	149	534	1,454	783
Observations	146,170	69,426	15,061	36,523	25,147
Panel B: Each adoption state					
$Adoption_t$, receiver only (CT)	−0.17	–	0.42*	–	−0.07
	(0.27)		(0.22)		(0.36)
$Adoption_t$, sender only (TC)	−1.00***	–	−0.49**	−0.97***	−0.85***
	(0.20)		(0.19)	(0.06)	(0.16)
$Adoption_t$, both (TT)	−0.58*	−0.64	−0.33**	–	−0.97***
	(0.34)	(0.69)	(0.13)		(0.22)
Mean pre Y	1.13	1.16	1.16	1.13	1.05
Wild-boot p, CT	0.525	–	0.154	–	0.832
Wild-boot p, TC	0.024	–	0.097	0.016	0.022
Wild-boot p, TT	0.298	0.648	0.142	–	0.036
Share of observations, CT	0.003	0.000	0.018	0.000	0.007
Share of observations, TC	0.145	0.000	0.011	0.374	0.284
Share of observations, TT	0.203	0.334	0.273	0.000	0.100
Dyads	2,842	149	534	1,454	783
Observations	146,170	69,426	15,061	36,523	25,147
Link effects	Yes	Yes	Yes	Yes	Yes
Month effects	Yes	Yes	Yes	Yes	Yes
Week-of-year effects	Yes	Yes	Yes	Yes	Yes
Directorate linear trends	Yes	Yes	Yes	Yes	Yes

Notes. This table reports the effect of adoption on processing time by the adoption state of the link. Processing time is the working days from the request for a document number to dispatch, and each observation is a document. Columns split documents by receiver, as in Table II, Panel A. Panel A treats a link once the sender has adopted (*TC* or *TT*), so links where only the receiver has adopted (*CT*) stay in the comparison group. Panel B estimates *CT*, *TC* and *TT* separately, with *CC* omitted. A dash marks a state with no documents in that column. Standard errors, clustered by link, in parentheses. p-values from a wild-cluster bootstrap (Webb) by regional directorate.

B. Communication

FIGURE A.4
DISTRIBUTION OF PROCESSING TIME, BEFORE AND AFTER ADOPTION



Notes. This figure shows the distribution of processing time before and after a link is treated. Processing time is the working days from the request for a document number to dispatch. Each bar is the share of documents with that processing time, and the last bar groups ten days or more. The sample is that of Table II, Panel A, column 1.

TABLE A.3
PROCESSING TIME BY SENDER DEPARTMENT

	(1) All	(2) Education and strategy	(3) Finance	(4) Other departments
$AdoptionTech_{st}$	-0.79***	-0.86***	-0.89***	-0.28*
	(0.24)	(0.30)	(0.32)	(0.16)
Mean pre Y	1.13	0.96	1.24	1.15
Effect, % of pre mean	-69.4	-88.9	-71.2	-24.3
Wild-boot p	0.070	0.004	0.103	0.296
Dyads	2842	964	471	1407
Observations	146170	22830	42518	80822

Notes. This table reports the effect of adoption on processing time by the department that sends the document. Processing time is the working days from the request for a document number to dispatch, and each observation is a document. Column 1 pools all departments and columns 2 to 4 are separate samples. The specification is that of Table II, Panel A, column 1. The pre-adoption mean is taken before either side of the link adopts. Standard errors, clustered by link, in parentheses. p-values from a wild-cluster bootstrap (Webb) by regional directorate.

TABLE A.4
DOCUMENTS AND ACTIVE LINKS AT THE DIRECTORATE LEVEL

	(1) Total documents	(2) Active links	(3) Documents per link	(4) Provision links	(5) Compliance links
$AdoptionTech_{st}$	-20.46	-1.07	0.11	3.77	-6.29
	(18.26)	(6.54)	(0.61)	(3.44)	(3.90)
Mean pre Y	143.90	58.16	3.30	4.62	9.83
Effect, % of pre mean	-14.2	-1.8	3.4	81.6	-64.1
Wild CI upper, % of pre	13.4	24.2	44.2	364.3	25.9
Wild-boot p (sender)	0.301	0.873	0.844	0.310	0.191
Observations	1346	1346	1346	1346	1346

Notes. This table reports the effect of adoption on the documents each directorate sends. Each observation is a sending directorate in a month. Column 1 counts documents, column 2 active links, those that carry at least one document in the month, and column 3 documents per active link. Columns 4 and 5 count active provision and compliance links. One row gives the upper bound of the wild-bootstrap confidence interval as a percent of the pre-adoption mean. All columns include directorate and month effects and directorate-specific linear trends. Standard errors, clustered by region, in parentheses. p-values from a wild-cluster bootstrap (Webb).

TABLE A.5
PROVISION AND COMPLIANCE DOCUMENTS PER LINK-WEEK

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Provision				**Compliance**			
	All link-weeks			Weeks with any	All link-weeks			Weeks with any
	Documents	Any document	Documents beyond the first	Documents	Documents	Any document	Documents beyond the first	Documents
Adoption	0.175***	0.106***	0.069**	0.394***	-0.026	0.041***	-0.067***	-0.633***
	(0.042)	(0.015)	(0.028)	(0.123)	(0.030)	(0.011)	(0.020)	(0.063)
Mean Y (pre)	0.046	0.039	0.007	1.167	0.277	0.162	0.115	1.712
Wild-boot p	0.000	0.000	0.000	0.004	0.395	0.000	0.001	0.000
Observations	81,237	81,237	81,237	3,795	77,106	77,106	77,106	12,266
Link effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Week-of-year effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Directorate linear trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. This table decomposes the effect of adoption on provision and compliance documents. Each observation is a sender department–receiver link in a week, with zeros in the weeks between the link's first and last document. Column 1 counts documents, column 2 indicates any document, and column 3 counts documents beyond the first, so columns 2 and 3 add up to column 1. Column 4 keeps weeks with at least one document and reproduces Table II, Panel B. Columns 5 to 8 repeat this for compliance documents. Standard errors, clustered by link, in parentheses. p-values from a wild-cluster bootstrap (Webb) by regional directorate.

C. Enrollment

Table A.6
Applications, raw and capacity-weighted

	(1) Applications	(2) First-time	(3) Repeat	(4) Applications (weighted)	(5) First-time (weighted)	(6) Repeat (weighted)
$AdoptionTech_{rt}$	4.17***	2.42**	1.75***	-1.26	-2.72	1.46***
	(0.73)	(0.88)	(0.40)	(1.58)	(1.62)	(0.22)
Wild-boot p	0.037	0.052	0.012	0.476	0.217	0.027
Mean Y (pre)	56.1	45.6	10.4	56.2	45.8	10.5
Observations	17725	17725	17725	17612	17612	17612

Notes. This table reports the effect of adoption on applications per center, split into first-time and repeat applicants. Each observation is a center in a year, 2016–2023. An application record is one per child and year, at the center where the child is registered. First-time applicants have no record in an earlier year, and repeat applicants are all others. Columns 1 to 3 weight every center equally, and columns 4 to 6 weight centers by their baseline size. All columns include center, year and macro-zone $\times$ year effects. Standard errors, clustered by regional directorate (16), in parentheses. p-values from a wild-cluster bootstrap (Webb) by regional directorate.

TABLE A.7
THE ENROLLMENT PIPELINE INSIDE AND OUTSIDE THE ENROLLMENT WINDOW, BY MANAGEMENT TYPE

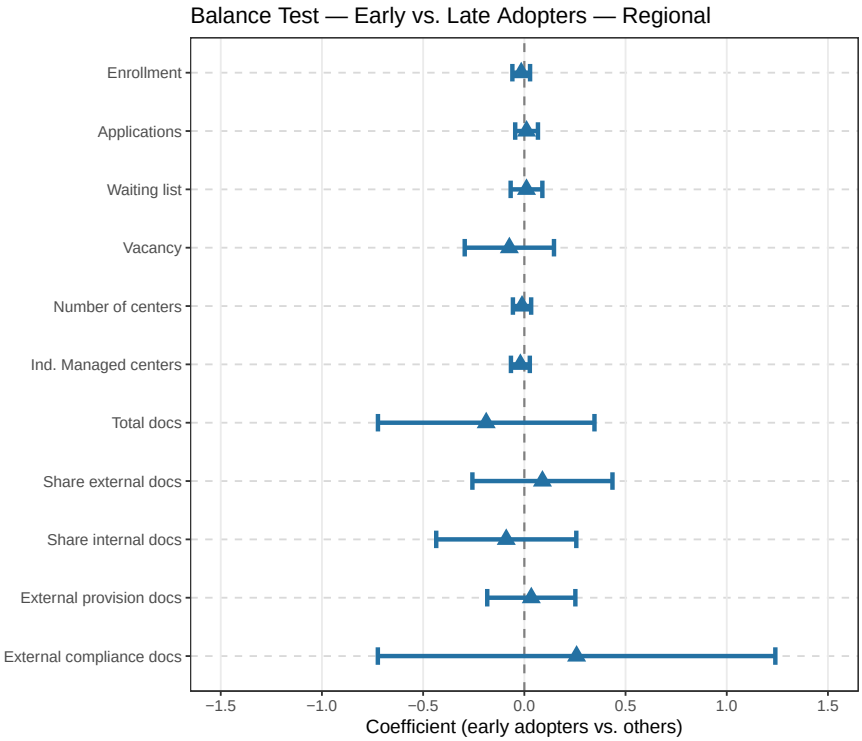
	(1) Applications	(2) Offers	(3) Acceptance rate	(4) Declined	(5) No answer	(6) Enrolled	(7) Waiting list
Panel A: Inside the window, indirectly managed centers							
$AdoptionTech_{rt}$	-0.92	3.18***	0.89	0.99***	-0.86***	3.05***	-4.14***
	(1.16)	(0.78)	(1.25)	(0.20)	(0.25)	(0.72)	(1.09)
Wild-boot p	0.583	0.027	0.641	0.046	0.085	0.025	0.066
Mean Y (pre)	36.3	27.7	90.8	1.3	1.4	25.0	8.5
Observations	12479	12479	12378	12479	12479	12479	12479
Panel B: Inside the window, directly managed centers							
$AdoptionTech_{rt}$	-0.22	-0.88	0.39	0.44	-1.65***	0.33	0.72
	(0.84)	(1.50)	(0.99)	(0.28)	(0.31)	(1.37)	(0.98)
Wild-boot p	0.887	0.633	0.747	0.234	0.021	0.853	0.617
Mean Y (pre)	61.8	48.4	87.8	2.8	3.1	42.4	13.1
Observations	5246	5246	5215	5246	5246	5246	5246
p: indirect = direct	0.000	0.001	0.296	0.001	0.000	0.003	0.992
Panel C: Outside the window, indirectly managed centers							
$AdoptionTech_{rt}$	5.61***	2.36***	5.17	0.01	0.11	2.25***	3.19***
	(1.58)	(0.67)	(5.81)	(0.07)	(0.07)	(0.56)	(0.91)
Wild-boot p	0.060	0.071	0.715	0.922	0.344	0.070	0.055
Mean Y (pre)	9.1	7.0	93.1	0.1	0.2	6.6	2.1
Observations	12479	12479	9286	12479	12479	12479	12479
Panel D: Outside the window, directly managed centers							
$AdoptionTech_{rt}$	1.60	0.35	15.40*	-0.12	-0.03	0.50	1.21
	(1.74)	(1.25)	(7.27)	(0.10)	(0.21)	(1.08)	(1.33)
Wild-boot p	0.443	0.855	0.128	0.390	0.918	0.807	0.492
Mean Y (pre)	21.5	16.3	91.7	0.5	0.8	15.1	5.2
Observations	5246	5246	4449	5246	5246	5246	5246
p: indirect = direct	0.000	0.001	0.793	0.001	0.000	0.001	0.001
Center effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Macro-zone × year effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. This table splits the enrollment pipeline of Table V by application date. Panels A and B use children who applied inside the enrollment window, October to March, and Panels C and D those who applied outside it. Each cell is a separate regression at the center-year level, 2016–2023. Offers are enrolled plus declined plus no answer, and applications are offers plus the waiting list, up to a few pending applications. The acceptance rate is enrolled over offers, in percent. The last row of Panels B and D tests that the indirect and direct coefficients are equal, in a regression that pools both types. Standard errors, clustered by regional directorate (16), in parentheses. p-values from a wild-cluster bootstrap (Webb) by regional directorate.

D. Threats to identification

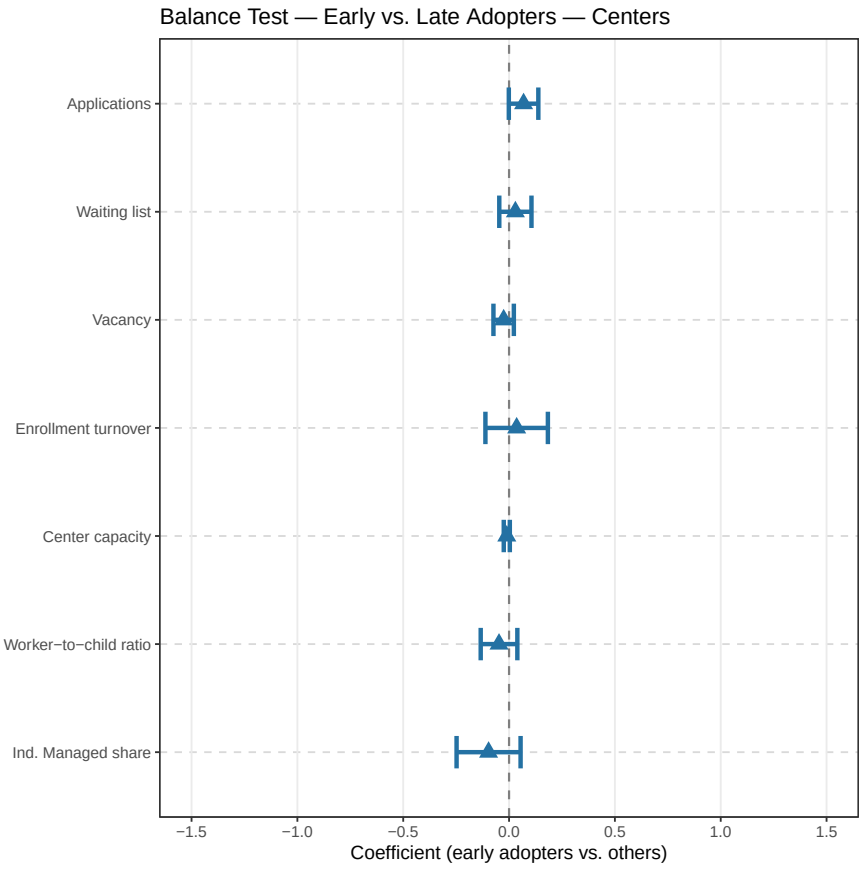
D.1 Pre-adoption trends.

FIGURE A.5
PRE-ADOPTION TRENDS, REGIONAL-DIRECTORATE VARIABLES



Notes. This figure shows pre-adoption trend differences between early and late adopting regional directorates. For each variable, the standardized year-to-year change is regressed on an early-adopter indicator and year effects, using pre-adoption years only. Each dot is a coefficient and each line its 95% confidence interval. The variables describe enrollment and communication at the regional directorate. Standard errors clustered by regional directorate.

Figure A.6
Pre-adoption trends, center variables



Notes. This figure repeats the test of Appendix Figure A.5 on center-level variables. Each dot is a coefficient and each line its 95% confidence interval. Standard errors clustered by regional directorate.

D.2 Pandemic overlap.

TABLE A.8
ADOPTION TIMING AND PANDEMIC CONDITIONS

	Adopted by t			
	(1) Lockdown days	(2) COVID visits	(3) Visits ages 1–4	(4) Share reopened
Panel A: All months				
X_t	-0.0016	0.0410	0.0523	0.0724
	(0.0066)	(0.0593)	(0.0498)	(0.1168)
Wild-boot p	0.816	0.555	0.366	0.545
Randomization p	0.804	0.476	0.321	0.545
ΔPr per SD (pp)	-0.67	17.61	14.39	2.41
Observations	912	912	912	240
Panel B: Pre-adoption months only				
X_t	-0.0016	0.0178	0.0167	-0.0115
	(0.0047)	(0.0168)	(0.0165)	(0.0928)
Wild-boot p	0.737	0.365	0.363	0.938
ΔPr per SD (pp)	-0.49	5.80	3.18	-0.40
Observations	734	734	734	61
Panel C: One lead and one lag				
X_{t-1}	-0.0050	0.0457	0.0420	0.1400
	(0.0052)	(0.0579)	(0.0516)	(0.1384)
X_t	0.0016	0.0157	0.0191	-0.0380
	(0.0035)	(0.0166)	(0.0133)	(0.0256)
X_{t+1}	0.0002	-0.0124	0.0045	-0.0032
	(0.0043)	(0.0269)	(0.0176)	(0.1085)
Wild-boot p	0.659	0.408	0.214	0.194
Sum of leads and lags	-0.0032	0.0490	0.0656	0.0988
p (sum)	0.718	0.500	0.309	0.590
Observations	880	880	880	208

Notes. This table reports regressions of adoption on pandemic conditions. Each observation is a regional directorate in a month, from January 2017 to September 2021. The outcome equals one from the month in which the directorate adopts the platform. Each column is a pandemic measure, lockdown days, emergency visits attributed to COVID (inverse hyperbolic sine), the same visits for ages one to four, and the share of days at Phase 3 or above of the *Paso a Paso* plan, which starts in July 2020. Panel A uses all months, Panel B pre-adoption months only, and Panel C adds one lead and one lag of each measure. All columns include directorate and month effects. Standard errors clustered by regional directorate (16). p-values from a wild-cluster bootstrap (Webb) and from 2,000 permutations of the adoption dates.

Table A.9
Pandemic exposure by adoption cohort

	Early (2020)	Late (2021)	Diff.	Norm. diff.	Exact p
Panel A: All sixteen directorates					
Lockdown days, 2020	27.09	45.55	-18.46	-0.59	0.215
Lockdown days, Mar–May 2020	2.38	8.70	-6.32	-0.91	0.038
Share of days at Phase 3+, 2020	0.80	0.72	0.08	0.49	0.315
Panel B: Excluding the Metropolitan region					
Lockdown days, 2020	27.09	28.57	-1.48	-0.09	0.889
Lockdown days, Mar–May 2020	2.38	4.93	-2.55	-0.64	0.226
Share of days at Phase 3+, 2020	0.80	0.80	-0.00	-0.04	0.958

Notes. This table compares pandemic exposure across the two adoption cohorts. Each observation is a regional directorate. Lockdown days are the mean across the region's municipalities of days under mandatory quarantine, and reopening is the share of days at Phase 3 or above. Columns give the cohort means, their difference, the normalized difference of Imbens and Rubin, and an exact p-value from every assignment of the cohort label across directorates. Panel B excludes the Metropolitan region, the only directorate whose 2020 lockdown exceeds the early-cohort maximum.

TABLE A.10
ENROLLMENT WITH PANDEMIC CONTROLS

	Enrollment		
	(1)	(2)	(3)
	Baseline	+ Lockdown	+ Lockdown, visits
Panel A: All sixteen directorates			
$AdoptionTech_{rt}$	5.67***	2.37*	2.96*
	(0.43)	(1.23)	(1.45)
Lockdown days		-0.05***	-0.02
		(0.01)	(0.02)
COVID visits (asinh)			-1.28***
			(0.27)
Wild-boot p	0.008	0.176	0.136
Mean Y (pre)	39.0	39.0	39.0
Observations	10290	10290	10290
Panel B: Excluding the Metropolitan region			
$AdoptionTech_{rt}$	5.52***	4.34***	6.58***
	(0.98)	(0.82)	(0.83)
Lockdown days		-0.04***	-0.01
		(0.01)	(0.01)
COVID visits (asinh)			-1.45***
			(0.28)
Wild-boot p	0.069	0.039	0.027
Mean Y (pre)	33.9	33.9	33.9
Observations	7622	7622	7622

Notes. This table adds pandemic controls to the enrollment specification. Each observation is a center in a year, 2016–2020, the window in which the late cohort is a not-yet-treated comparison. Column 1 is the baseline, column 2 adds lockdown days, and column 3 adds COVID emergency visits (inverse hyperbolic sine). Panel B excludes the Metropolitan region. All columns include center, year and macro-zone $\times$ year effects. Standard errors, clustered by regional directorate (16), in parentheses. p-values from a wild-cluster bootstrap (Webb) by regional directorate.

TABLE A.11
ENROLLMENT AT PROVIDERS THE REFORM DID NOT REACH

	Agency centers		Providers outside the agency	
	(1)	(2)	(3)	(4)
Panel A: All sixteen directorates				
$AdoptionTech_{rt}$	5.67***	2.37*	0.10	-0.09
	(0.43)	(1.23)	(0.18)	(0.27)
Lockdown days		-0.05***		-0.00
		(0.01)		(0.00)
Wild-boot p	0.009	0.178	0.575	0.682
Mean Y (pre)	39.0	39.0	48.4	48.4
Observations	10290	10290	15638	15638
Panel B: Excluding the Metropolitan region				
$AdoptionTech_{rt}$	5.52***	4.34***	0.50	0.32
	(0.98)	(0.82)	(0.52)	(0.55)
Lockdown days		-0.04***		-0.01
		(0.01)		(0.00)
Wild-boot p	0.061	0.036	0.331	0.555
Mean Y (pre)	33.9	33.9	46.5	46.5
Observations	7622	7622	10940	10940

Notes. This table repeats the enrollment specification for providers the reform did not reach. Columns 1 and 2 are the agency's centers and columns 3 and 4 all providers outside the agency, non-profit and private. For these providers, the adoption of their regional directorate is a placebo. In each pair, the second column adds lockdown days. The window is 2016–2020, as in Appendix Table A.10, and Panel B excludes the Metropolitan region. Agency centers come from the application records and other providers from the center panel, so pre-adoption means differ across groups. All columns include center, year and macro-zone $\times$ year effects. Standard errors, clustered by regional directorate (16), in parentheses. p-values from a wild-cluster bootstrap (Webb) by regional directorate.

D.3 Regional shocks.

TABLE A.12
ENROLLMENT EFFECT UNDER REGIONAL CONTROLS

	Enrollment per center				
	(1) Year effects	(2) Main	(3) Directorate trends	(4) Child population	(5) Lockdown × year
$AdoptionTech_{rt}$	3.99	4.17***	4.04***	4.18***	3.95**
	(2.45)	(0.42)	(0.47)	(0.43)	(1.67)
Wild-boot p	0.419	0.014	0.024	0.015	0.227
Mean Y (pre)	38.9	38.9	38.9	38.9	38.9
Observations	17,725	17,725	17,725	17,724	17,725
Macrozone × year effects	No	Yes	Yes	Yes	Yes
Directorate linear trends	No	No	Yes	No	No
Child population	No	No	No	Yes	No
Lockdown exposure × year	No	No	No	No	Yes
Center effects	Yes	Yes	Yes	Yes	Yes
Year effects	Yes	Yes	Yes	Yes	Yes

Notes. This table reports the effect of adoption on enrollment per center under regional controls, 2016–2023. Column 2 is the main specification of Table III. Column 1 drops the macro-zone × year effects. Columns 3 to 5 add, one at a time, directorate-specific linear trends, the child population of the center's municipality, and the lockdown days of the region from March to May 2020 interacted with year indicators. Standard errors, clustered by regional directorate (16), in parentheses. p-values from a wild-cluster bootstrap (Webb) by regional directorate.

D.4 Staggered adoption.

TABLE A.13
COMMUNICATION EFFECTS UNDER HETEROGENEITY-ROBUST ESTIMATORS

	(1)	(2)	(3)	(4)
	Processing time (days)	Documents sent per week		
		Total	Provision	Compliance
TWFE	−0.79	−0.24	0.39	−0.63
	(0.24)	(0.21)	(0.12)	(0.06)
Wild-boot p	0.070	0.335	0.032	0.049
Treated obs. with negative weight (%)	48.6	49.3	47.4	37.0
Treated obs. with zero weight (%)	0.2	0.0	0.0	0.0
Imputation (Borusyak et al.)	−0.88	−0.49	0.73	−0.35
	(0.40)	(0.37)	(0.30)	(0.14)
Bootstrap p	0.028	0.189	0.015	0.014
Treated obs. imputed (%)	28.2	27.6	9.6	29.5
Mean Y (pre)	1.14	2.50	1.17	1.71
Observations	146,170	60,914	3,795	12,266

Notes. This table compares the two-way fixed-effects (TWFE) estimates of Table II with the imputation estimator of Borusyak *et al.* (2024). Column 1 is processing time, from Panel A, and columns 2 to 4 are documents per week, from Panel B. The weight rows give the share of treated observations with a negative or zero weight in the TWFE estimate (de Chaisemartin & D'Haultfœuille, 2020). The imputation estimator fits the fixed effects on untreated observations only and predicts the untreated outcome for treated ones. The imputed row gives the share of treated observations for which this prediction is possible. TWFE inference as in Table II. Imputation p-values from a bootstrap that resamples links.

TABLE A.14
ENROLLMENT EFFECTS UNDER HETEROGENEITY-ROBUST ESTIMATORS

	(1) All centers	(2) Indirectly managed	(3) Directly managed
TWFE	4.17	5.30	0.83
	(0.42)	(1.04)	(2.19)
Wild-boot p	0.014	0.028	0.817
Treated obs. with negative weight (%)	22.8	20.9	26.1
Treated obs. with zero weight (%)	32.4	35.6	26.6
Imputation (Borusyak et al.)	6.15	7.05	2.69
	(1.05)	(3.26)	(5.77)
Bootstrap p	0.000	0.031	0.641
Treated obs. imputed (%)	6.8	6.7	6.8
Callaway–Sant'Anna	4.30	4.40	2.97
	(3.34)	(3.89)	(2.15)
p	0.198	0.258	0.167
Mean Y (pre)	38.91	31.63	57.50
Observations	17,725	12,479	5,246

Notes. This table compares the two-way fixed-effects (TWFE) estimates of enrollment in Table III, column 1, and Table IV, columns 1 and 2, with the imputation estimator of Borusyak *et al.* (2024) and the estimator of Callaway & Sant'Anna (2021). Each observation is a center in a year, 2016–2023. The weight rows give the share of treated observations with a negative or zero weight in the TWFE estimate (de Chaisemartin & D'Haultfœuille, 2020). The imputation estimator fits the fixed effects on untreated center-years only, and its p-values come from a bootstrap that resamples regional directorates. The Callaway–Sant'Anna estimate averages the group-time effects, with not-yet-treated centers as controls. TWFE inference as in Table III.

E. Alternative explanations

TABLE A.15
WRITTEN QUERIES FROM FAMILIES TO THE DIRECTORATE

	ln(queries + 1)			
	All directorates		Excluding the national directorate	
	(1)	(2)	(3)	(4)
	Year	By enrollment window	Year	By enrollment window
AdoptionTech$_{rt}$	0.068	0.083	-0.117	-0.077
	(0.207)	(0.195)	(0.102)	(0.136)
× in-window		-0.033		-0.087
		(0.153)		(0.163)
Wild-boot p (Adoption)	0.874	0.728	0.266	0.573
Wild-boot p (× in-window)		0.835		0.621
Mean Y (pre, count)	150.9	150.9	155.6	155.6
Offices	17	17	16	16
Observations	816	816	768	768

Notes. This table reports the effect of adoption on the written queries that families send to the directorate. Each observation is a directorate in a month, 2020–2023, and the outcome is ln(queries + 1). Columns 1 and 3 report the effect over the year, and columns 2 and 4 add its interaction with the enrollment window. Columns 3 and 4 exclude the national directorate. The registry has no query type, so the count includes every written query a directorate receives. All columns include directorate and month effects. Standard errors, clustered by directorate, in parentheses. p-values from a wild-cluster bootstrap (Webb). Source: SAIP written-query registry.

TABLE A.16

THE PAYMENT RULE AND THE ENROLLMENT EFFECT AT INDIRECTLY MANAGED CENTERS

Panel A: Transfers and enrollment

	Change in log transfers
Change in log enrollment	0.293***
	(0.036)
× Transfers computed on 2019 attendance (2020–22)	−0.289***
	(0.040)
Elasticity under the 2019-attendance rule (sum)	0.004
	(0.015)
Wild-boot p (interaction)	0.001
Center-year differences	8,201

Panel B: Enrollment effect, by estimation window

	(1) 2016–20	(2) 2016–22	(3) 2016–23
$AdoptionTech_{rt}$	5.68***	5.35***	5.30***
	(1.08)	(1.05)	(1.04)
Wild-boot p	0.027	0.026	0.028
Mean Y (pre)	31.7	31.6	31.6
Centers	1,643	1,689	1,690
Observations	7,391	10,792	12,479

Notes. This table reports how transfers respond to enrollment at indirectly managed centers and the enrollment effect at those centers over three windows. Transfers are a value per child times the center's average monthly attendance, summed over the year. From 2020 to 2022 the agency computed them on each center's average attendance in 2019 (JUNJI, 2022c). Panel A regresses the change in log transfers on the change in log enrollment between consecutive years, with year effects, and interacts it with the years of the 2019-attendance rule. Panel B estimates the specification of Table IV, column 1, on each window. Standard errors, clustered by regional directorate (16), in parentheses. p-values from a wild-cluster bootstrap (Webb) by regional directorate.

F. Costs

TABLE A.17
PERSONNEL SPENDING AND ENROLLMENT AT INDIRECTLY MANAGED CENTERS

	(1) Enrolled children	(2) Personnel (CLP mn)	(3) Child-scaling inputs (CLP mn)	(4) Total (CLP mn)	(5) Personnel per child (CLP th.)	(6) Log personnel
$AdoptionTech_{rt}$	4.511***	6.634***	-0.505	5.809***	-263.581	-0.011
	(1.228)	(2.039)	(0.811)	(1.743)	(484.174)	(0.011)
Wild-boot p	0.040	0.117	0.738	0.066	0.735	0.360
95% interval, lower	0.657	-5.636	-2.545	-3.301	-3.1e+03	-0.071
95% interval, upper	10.008	15.552	2.444	13.795	708.572	0.031
Implied by one-for-one hiring		12.688	0.960	14.472	0.000	0.136
Implied by fixed staff		0.000	0.000	0.959	-578.182	0.000
Pre-adoption mean	30.968	87.104	6.589	99.347	4547.277	18.137
Center-years	8,338	8,338	8,338	8,338	8,309	8,326
Centers	1,688	1,688	1,688	1,688	1,688	1,688

Notes. This table reports the effect of adoption on enrollment and spending at indirectly managed centers that file expenditure returns with the agency. Each observation is a center in a year, 2019–2023. Column 1 is enrolled children. Columns 2 to 5 are accepted spending on personnel (wages, fees and severance), on the inputs that scale with a child, in total, and on personnel per enrolled child. Column 6 is column 2 in logs. Two rows give the change implied if staff grow in proportion to enrollment, using the effect in column 1, and if staff stay fixed. The specification is that of Table III. Standard errors, clustered by regional directorate (16), in parentheses. p-values and 95% intervals from a wild-cluster bootstrap (Webb) by regional directorate.

G. Data and measurement

Sample of centers. The sample covers the agency's traditional childcare centers, the *jardín infantil* modality, under both management types. It excludes alternative modalities and the *Transición* I and II levels, which serve children older than four and are often part of elementary schools. Primary and secondary education involve multiple agencies and levels within the same provider, which would confound the link between communication and delivery. The enrollment panel covers 2016 to 2023 and contains the 2,480 centers of Table I, 790 directly and 1,690 indirectly managed. Indirectly managed centers are those of the transfer program (*VTF*), run by municipalities and non-profit operators.

Applications and enrollment. The enrollment system records one application per child and year, with its filing date, the center applied to and its final status at the close of the cycle. A child is enrolled, offered a seat without answering, declined after an offer, or on the waiting list. Offers are the sum of the first three.

Enrollment year. I assign each application to the calendar year in which it is filed, with its final status. Year t therefore contains the applications filed from October to December of t, which open the cycle of the next school year, and those filed from January to September of t. Applications filed from October to March are inside the enrollment window, and those filed from April to September are outside it.

Household income. The *Registro Social de Hogares* ranks households by an estimate of permanent income. I use the bottom 60%, the cutoff of the agency's priority model and the group reported in Table VII, Panel B.

Communications. Each record is a document sent by the agency, with its sender department, its receiving entity and the dates of the request for a document number and of dispatch. Processing time is the working days between the two. A link is a pair of a sender department and a receiving entity. Weekly outcomes count the documents a link sends in a week. The estimation sample keeps 15 quarters before and after the first adoption on each link. It contains 146,170 documents, sent by 119 departments of the 17 directorates to 923 receiving entities, on 2,842 links. The adoption date is the date onboarding with the Digital Government Division ends (§2.4).

Pandemic measures. Quarantine days are the mean across the region's municipalities of days under mandatory quarantine, from the Ministry of Health's municipal records as compiled in the Ministry of Science's COVID-19 repository. Through November 2020 they come from quarantine episodes with start and end dates, and from December 2020 from days at Phase 1 of the *Paso a Paso* plan. Emergency visits are the weekly emergency-care records by municipality and cause of the Ministry of Health (DEIS, datos.gob.cl), aggregated to the region and month, for all ages and for ages one to four. Reopening is the share of days in the month at Phase 3 or above of the *Paso a Paso* plan, which starts in July 2020.

Classifying documents by task. The registry records the sender department and the receiving entity of each document, not its content. Each document is therefore classified by the formal role of the sender department toward its receiver, as set out in the agency's organizational resolution (JUNJI, 2021a). Departments are grouped by function, and

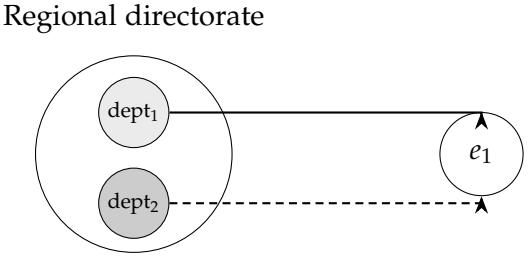
Appendix Table A.18 gives the rule. Documents to a department of the same directorate are internal. The rest are classified as follows. *Provision* is sent by the education quality and planning departments to the entities the enrollment process involves. *Education–other* is sent by the same departments to all other receivers. *Compliance* is sent by the finance department to the operators of indirectly managed centers. The other classes hold everything else. The rule uses no information on content or outcomes. Appendix Figure A.7 illustrates it.

TABLE A.18
CLASSIFICATION OF DOCUMENTS BY SENDER DEPARTMENT AND RECEIVER

Class	Sender department	Receiver
Internal	Any	Department of the same directorate
Provision	Planning	Other directorates of the agency
	Education quality	Central government and public services
	Education quality	Operators of indirectly managed centers and directly managed centers
Education–other	Education quality, planning	All other receivers
Compliance	Finance	Operators of indirectly managed centers
Other	Finance	All other receivers
	Executive direction	Any
	Human resources	Any
	Infrastructure, risk prevention and coverage	Any
	IT, legal, audit and records office	Any

Notes. This table reports the rule that assigns each document to a class. Each document is assigned by the function of its sender department, as set out in the agency's organizational resolution (JUNJI, 2021a), and by the type of its receiver. Departments are grouped by function across directorates. Documents with no recorded department fall in Other.

FIGURE A.7
CLASSIFICATION RULE



Notes. This figure illustrates the classification rule. Two departments of the same regional directorate send documents to the same entity e_1. Each document is classified by the function of its sender department toward e_1, so the solid and dashed documents can fall in different classes although sender directorate and receiver are the same.